Computers & Industrial Engineering 185 (2023) 109657

Contents lists available at ScienceDirect

Computers & Industrial Engineering

journal homepage: www.elsevier.com/locate/caie

Optimization of system resilience in urban rail systems: Train rescheduling considering congestions of stations

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ARTICLE INFO

Keywords:
Congestion of stations
Resilience improvement
Train rescheduling
Two-phase iterative optimization

ABSTRACT

In urban rail systems, unexpected events, due to equipment breakdown, extreme weather, etc., frequently cause the delay of trains and overcrowding of platforms, significantly affecting the service quality to passengers. Therefore, improving system resilience to recover to normal conditions more quickly has become a critical problem in the management of urban rail systems. In contrast to most previous studies that focused only on train rescheduling, our paper proposes optimizing the resilience of urban rail systems by considering the influence of delayed trains and overcrowded passenger flows. Specifically, we define a resilience-oriented quantitative evaluation index. Using the evaluation index as the objective function, we construct a mixed-integer nonlinear programming model. In our formulation, we consider a practical case in which the boarding and alighting of passengers at overcrowded stations may further delay trains. To solve the model, we develop a two-phase iterative optimization approach, in which phase 1 focuses on the adjustment of trains to enhance system resilience, and phase 2 evaluates the impact of passengers on the adjusted train schedule. Through the comparison of three different rescheduling methods, we prove the validity of the derived inequalities and our approach balances the computational time and system performance. Case studies are conducted on the Beijing Metro Line 1 to verify the effectiveness of the proposed approach. The results demonstrate that system resilience can be improved by as much as 39.7% using our optimization approach.

1. Introduction

Urban rail transit is a key solution to support mobility in high-density urban areas. As a mode of shared transportation, urban rail transportation systems carry large numbers of commuters in a more environmentally friendly manner than private transportation systems. Dependence on urban rail transit continues to increase in many cities worldwide. For example, in Beijing, a metropolis with 20 million people, more than 8 million metro trips were conducted daily in 2021, reaching a daily average of more than 10 million commuter passengers. However, as a critical public infrastructure, transportation systems are in a high-risk scenario and subject to unexpected disruptions from various influencing factors, including failures of system components, malicious damage, and extreme weather (Adjetey-Bahun, Birregah, Châtelet, et al., 2016). According to historical data from the Beijing Metro, unexpected failures occur once a day on average. In particular, nearly half of all disruptions are caused by the breakdown of signaling systems (or a subsystem of the signaling system), and such disruptions can last from two minutes to over an hour (Yin, Ren, Liu, et al., 2022). Disruptions in urban rail systems always lead to the infeasibility of planned timetables, causing delays or cancellations of a large number of trains (Ortega, Pozo, & Puerto, 2018). The frequent occurrence of unexpected events places considerable pressure on rail managers and causes overcongestion at key stations, which may even encourage the spread of epidemic diseases (Centers for Disease Control and Prevention (CDC), 2021). For example, on July 5, 2019, a total of 13 trains were canceled owing to a five-minute failure of the grid network on Beijing Metro Line 5 during morning peak hours, and the delay of trains finally recovered after 30 min (Yin, Ren, Su, et al., 2023b). Short-term operational disturbances (STODs), such as equipment failures and large passenger flows, which do not have significant consequences, occur more frequently than long-duration disturbances, such as natural disasters and terrorist attacks, which have significant negative effects (Yin et al., 2022). Although STODs do not cause significant damage to urban rail transit systems, their negative effects on the operational service level (OSL) of a system cannot be ignored. Therefore, to improve a

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https://doi.org/10.1016/j.cie.2023.109657

Available online 29 September 2023

system's OSL in daily operations, it is necessary to assess its ability to respond to STODs (Chen, Liu, Du and et al., 2022).

Resilience refers to the ability of a system to withstand major disruptions within an acceptable reduction in service performance and adapt to rapidly recover from perturbations. Recently, the concept of resilience has been successfully applied in many engineering fields (Dong & Frangopol, 2015; Elmqvist, Folke, Nyström, et al., 2003; Henry & Ramirez-Marquez, 2012; Zhang, Du, Huang, et al., 2018). As urban rail networks become denser and more complex, disruptions to urban rail systems are uncertain and inevitable. Therefore, instead of focusing on the prevention of disruptions, an increasing number of railway managers in recent years have shifted their attention to the ability to withstand and recover quickly from these disruptions (Yin et al., 2022). Based on the advantages in reflecting a system's ability to respond to disturbances, resilience measures provide new perspectives for assessing an urban rail system's responsiveness. In 2011, Cox, Prager, and Rose (2011) conducted the first study to assess the resilience of an urban rail system based on the losses of passenger trips and ticket fares during the July 7th London terrorist attacks. Subsequently, several researchers have studied the resilience of urban rail systems (Zhou, Wang, & Yang, 2019). Existing studies on the resilience assessment of urban rail systems can be divided into two categories according to the duration of disturbances. The first category emphasizes the capability to respond to catastrophic disturbances with long duration (e.g., severe natural disasters Jiao, Zhu, Huo, Wu, & Zhang, 2023; Qi, Meng, Zhao, et al., 2022; Yumagulova, 2020 and terrorist attacks Chen, Liu, Peng and et al., 2022). Moreover, Zhang, Chai, and Guo (2023) studied the characteristics (vulnerability, robustness, rapidity, and degree to return) of the metro system's responses to the disturbance with a multivariate multiple regression model and quantitatively evaluated COVID-19-caused resilience performance of metro rails using the metro transit data from the United States. Focusing on network topology characteristics, most studies measure the time-invariant performance of networks in terms of the network topology and/or transport functions. However, describing passenger travel services and time-varying characteristics of urban rail systems at the network topology level is often difficult, which creates a gap in the real-time rail traffic management of daily operations. The second research category, which assesses resilience under STODs that occur frequently in daily operations is limited. Chen, Liu, Du et al. (2022) asserted that assessing the resilience of urban rail systems to STODs is critical for improving their OSL during daily operations in terms of the high frequency of STOD occurrence. To the best of our knowledge, no study has been conducted on the optimization of the resilience of urban rail systems to STODs with consideration to the effects of overcrowding and limited train capacity. With the increase in public demand for urban rail transportation, a phenomenon in route choice called "left behind" has emerged, where passengers fail to board the first arriving train because of its high load (Oliker & Bekhor, 2020). Typically, when rescheduling trains, dispatchers tend to be concerned only with railway operation indicators that have the larger impact on their performance evaluation, including the punctuality, cancellations, whereas passengers "left behind" at platforms are often ignored. To simplify modeling problems, previous studies assumed that the capacity of the vehicle is sufficiently large such that all waiting passengers on the platform can board within the limited dwelling time. However, this assumption deviates from reality, particularly when disturbances occur during rush hour, where stranded passengers cause overcrowding of platforms, which is a common and urgent problem.

To bridge the aforementioned gaps and relieve pressure on stations during peak hours, this study developed a resilience-oriented train rescheduling model for an urban rail network under STODs that jointly considers the effect of station congestion. The contribution of this study is twofold. First, this paper assessed resilience of an urban rail system under STODs. A resilience-oriented quantitative evaluation index, which considers congestion level of stations, is proposed to measure the system performance. Second, this paper developed a two-phase iterative optimization approach for the resilience enhancement of urban rail systems to solve the proposed mixed-integer nonlinear programming (MINLP) model. In the small case study, the effectiveness of our proposed approach is demonstrated in terms of acceptable computational time and more outstanding resilience performance, and further studies are conducted on different adjustment strategies and congestion levels through the real-world case based on the operational data of Beijing Metro Line 1. The results demonstrate that the proposed approach can be used by decision-makers and urban railway operators to adopt an efficient rescheduling strategy in time in case of disruptions. The outcomes also contribute to the capacity design of key stations, in particular for transfer stations.

The remainder of this paper is organized as follows: Section 2 reviews the relevant literature on disruption management and quantitative assessment of resilience in rail transit systems. This problem is described in detail in Section 3. In addition, the formulation of our MINLP is explicitly described, including the notations, objective functions and constraints, in Section 4. Section 5 proposes a two-phase iterative optimization approach for the MINLP model. Section 6 presents case studies and analyzes the numerical results. Finally, we conclude this paper in Section 7.

2. Literature review

For a large and complex system such as an urban rail network, functionality can be crippled by interruptions or even smaller perturbations, leading to train delays and a large number of stranded passengers at rail stations. Under disruptions, rail dispatchers adjust train schedules to reduce the negative effects on passengers. Nevertheless, the decisions manually made by dispatchers lack rigorous computation or optimization, which frequently take a long time to recover to normal conditions. Therefore, an increasing number of recent studies have focused on the management of disruptions and the enhancement of resilience in urban rail systems. Next, we review the relevant literature on (i) disruption management and (ii) resilience quantification and optimization in urban rail transit systems.

2.1. Disruption management in rail transit systems

In urban rail systems, daily rail transit operations are often disrupted by various accidents, such as infrastructure or rolling stock failures. Disruption management, considered one of the most challenging tasks in railways, responds to such events by adjusting train schedules and rolling stock circulation plans. Real-time train timetable rescheduling and rolling stock rescheduling are recognized as two key steps that must be performed in case of disruptions (Cacchiani, Huisman, Kidd, et al., 2014).

The train timetable provides passengers with the exact arrival and departure times of trains at each station. In normal operations, the planned timetable is conflict-free, i.e., only one train is permitted to occupy the same block section or minimum headway distance (Ariano, Pacciarelli, & Pranzo, 2007). However, under disruptions, the planned timetable must be rescheduled in real time to adjust the arrival/departure times of trains and the routes trains take at stations, or even to cancel some trains owing to the disruption of some sections or stations. The aim is to regenerate a conflict-free train timetable with adequate service for traveling passengers, i.e., to minimize the delay time and number of cancellations (Corman, 2010). In this paper, approaches to train rescheduling are divided into operation-centric and passenger-centric models.

From an operation-centric perspective, Brucker, Heitmann, and Knust (2002) considered the problem of rescheduling trains in the event of track closure owing to construction works on a double-track line. A local search heuristic with minimum lateness was presented and tested on a real-world instance. Based on the integer linear programming (ILP) model proposed by Fioole, Kroon, Maróti, and Schrijver (2006), Nielsen, Kroon, and Maróti (2012) addressed the rolling stock redistribution after disruptions using a rolling horizon approach. The generic framework was successfully applied to specific realistic rolling stock rescheduling problems of the Netherlands Railways (NS). Narayanaswami and Rangaraj (2013) developed a mixed-integer linear programming (MILP) model to resolve conflicts caused by disruptions that block parts of a single bidirectional line. Train movements are rescheduled in both directions of the line for a small artificial instance to minimize the total delays of all trains. Louwerse and Huisman (2014) considered partial and complete blockades under a major disruption on a double track line. They developed a mixed-integer programming model for generating a disposition timetable with two rescheduling possibilities for trains: cancellations and delays. Zhan, Kroon, Zhao, et al. (2016) focused on the rescheduling of high-speed trains on a double-track line with a partial blockade (unavailability of one track). They formulated the investigated problem as an MILP and used a rolling-horizon approach to solve it. The model was tested on the Beijing–Shanghai high-speed railway line. Similarly, using the rolling-horizon algorithm, Peng, Yang, Ding, et al. (2023) designed an MILP model integrated the real-time train rescheduling and the train speed management in consideration of the disruption uncertainties.

From the passenger-centric perspective, Cadarso, Marín, and Maróti (2013) proposed an integrated optimization model for timetable and rolling stock rescheduling that considers dynamic passenger demand. The timetabling and rolling stock rescheduling problem was formulated and solved as an MILP model subject to the anticipated demand calculated using a logit model. In their study, recovery strategies included canceling existing train services or scheduling additional services. However, the possibility of retiming existing trains was not considered. Kroon, Maróti, and Nielsen (2015) presented a mathematical model and iterative heuristic to solve the real-time rolling stock rescheduling problem with dynamic passenger flows. The model minimizes a combination of system-related costs (such as penalties for the modification of rolling stock compositions) and service-related costs that express the effect of train capacity on the total passenger delay. Gao, Kroon, Schmidt, and Yang (2016) addressed the rescheduling problem of metro lines after disruptions with time-varying passenger demand. Their study observed that if the skip-stop strategy is properly adjusted during the recovery stage, the total waiting time of passengers can be significantly reduced. By employing a space–time network representation, Yin, Yang, Tang, et al. (2017) rigorously formulated a complex train scheduling and control problem into two MILP models to jointly minimize the total passenger waiting time and energy consumption. Moreover, an efficient Lagrangian relaxation (LR)-based heuristic algorithm was developed to obtain a solution within a short computational time. Wang, Ariano, Yin, et al. (2018) proposed an integrated optimization model for train timetabling and rolling stock circulation planning for urban rail lines, in which the train departure headways between train services were optimized according to time-varying passenger arrival rates. Yin, Pu, Yang, D'Ariano, and Wang (2023a) further extended the integrated optimization model to the application scenario with multiple lines and rolling stock depots and developed a Benders decomposition-based solution algorithm that can improve the service quality by 17.6%, on average, using the same fleet size of rolling stock.

In recent years, disruption management has considered complex real-world scenarios. Considering two recovery strategies, namely, alternating train directions and permitting short turns, Huang, Mannino, Yang, et al. (2020) coupled big-M and time-indexed formulations to linearize a proposed MINLP model for real-time applications. Zhu and Goverde (2021) developed the sequential and combined approaches to reschedule the timetable in the event of multiple disruptions that occur at different geographic locations but have overlapping periods and are pairwise connected by at least one train line. To address long multiple connected disruptions more efficiently, they also proposed a new rolling-horizon method that can generate high-quality rescheduling solutions within an acceptable time. Focusing on the control of commuter flow, Xu and Ng (2020) formulated a robust optimization model to simulate commuter flow logic and evaluated the impact of uncertain disruptions by solving a max–min optimization problem. This problem can be reduced to an MILP model with moderate dimensions, which can be accomplished efficiently by applying commercial solvers such as CPLEX. Tang, Xu, Luo, et al. (2021) further developed and extended this model by explicitly considering commuter transfer walking times between different platforms.

2.2. Resilience quantification and assessment in rail transit systems

The concept of resilience was first introduced into ecological research in the early 1970s and was defined by Holling (1973) as the persistence of species in an ecological system and the relationships between the considered species facing man-made disturbances such as human domestic and industrial wastes and changes in fish populations by harvesting. Resilience has been widely applied in economics, sociology, engineering, and other fields. Transportation systems are at the forefront of various environmental and socioeconomic shocks; thus, the study of resilience has gained wide interest from scholars in the transportation field.

The resilience and vulnerability analysis on general transportation networks, particularly road networks with well-established measures, have been extensively studied in the existing literature (El Rashidy & Grant-Muller, 2019; Faturechi & Miller-Hooks, 2015; Gu, Fu, Liu, Xu, & Chen, 2020). In contrast to road transportation, rail transit networks could be more vulnerable but less resilient owing to fewer route alternatives and a large number of affected passengers, as well as the associated risks, such as passenger crowd tramples during incidents. Rail transit systems also include interdependent electrical, communication, and rail facility systems, which contribute to high exposure and vulnerability to incidents and disruptions (Lu, 2018). Ilalokhoin, Pant, and Hall (2023) evaluated the resilience of the southern region of Great Britain's rail network in scenarios of failure initiated in the traction power supply system that could disrupt 75% of all trains in the southern region and 25% of all trains nationally. Additionally, the network's vulnerability to failure at peak travel times with a delay during peak travel results in about 63% increase in passenger delay minutes compared with off-peak travel times for the same duration. As the vast literature indicates, most current research focuses on the implementation of complex network theory to quantify resilience of urban rail systems. These studies transform the rail transit line network into a directed graph and then analyze the system performance variations under certain edge failure conditions, which is vital to the planning and design of urban rail transit networks. For example, Reggiani (2013) highlighted the role of topological connectivity in the analysis of network resilience and outlined operational measures for enhancing network resilience. Cats, Koppenol, and Warnier (2017) measured the link criticality and degradation rapidity in a public transport network robustness model connecting local capacity reductions to network-wide performance changes. By utilizing the Washington, DC Metro as a case study, Saadat, Ayyub, Zhang, et al. (2020) observed that topology enhancement such as adding a loop line to the network can improve the resilience of the network prior to failure and significantly reduce the vulnerability of the network. Considering budget limitations and the uncertainty in repair time, Zhang, Ren, and Song (2022) introduced a resilience-based optimization model. This model aims to select an optimal restoration sequence scheme, which maximizes the overall average efficiency, under the condition that the network accessibility meets given resilience requirements. However, reflecting a series of industry characteristics, such as complex emergencies, multi-service linkages, and passenger-oriented services in rail transportation at the network topology level is difficult; these studies frequently ignore the impact on passengers, particularly in the context of large passenger

Table 1
Summary of relevant studies on disruption management and resilience analysis in rail transit systems.

Publications	Modeling approach	Objective function	Resilience metric	Capacity constraints		Solution method	Case city
				Train	Depot		
Kroon et al. (2015)	MILP	The total passenger delay and the system-related criteria	No	Yes	Yes	Heuristic iterative algorithm	Netherlands
Gao et al. (2016)	MINLP	The travel time of the considered services and the passenger flow	No	Yes	No	Heuristic iterative algorithm	Beijing
Huang et al. (2020)	MILP	The number of nonboarding passengers and headway deviation	No	Yes	No	Two-stage approach with CPLEX	Beijing
Jin et al. (2014)	MINLP	The expectation of the demand fulfillment	The fraction of travel demand that can be satisfied	No	No	Two-stage stochastic programme	Singapore
Lu (2018)	Graph model	The importance-impedance-based performance	The speed of network recovering from the worst performance to its original state	No	No	Resilience assessment	Shanghai
Tang et al. (2021)	MILP	The average commuter flows detaining	The ratio of remaining performance	Yes	No	Resilience assessment	Singapore, Chongqing
Shahabi et al. (2022)	Discrete-event simulation	Resiliency	The headway regularity index	Yes	No	Iterated local search	Karaj
Chen, Liu, Du et al. (2022)	Simulation-based model	The reciprocal of the average travel time of passengers	The ratio of average performance loss	No	No	Resilience assessment	Chengdu
Chen, Liu, Peng et al. (2022)	Bi-level programming model	Resiliency	The reciprocal of global accessibility (GA) indicator and proportion of unaffected passengers	No	No	Simulated annealing algorithm	Chengdu
Yin et al. (2023b)	MILP	Resiliency and extra cost of reserve rolling stocks	The train delay and canceled events	No	Yes	Branch-and-cut approach	Beijing
This paper	MINLP	Resiliency	The congestion level of the busiest station	Yes	Yes	Two-phase iterative approach	Beijing

flows in congested urban rail systems. Chen, Liu, Peng et al. (2022) developed a bi-level programming model that aims to maximize the global accessibility and global efficiency of an urban rail transit network considering the travel characteristics of passengers. Their proposed method can aid urban rail network planners in selecting an optimal solution when planning a new network. However, it cannot provide a real-time strategy for short-term disturbances in daily operations.

To model the resilience of rail transit systems more appropriately, a few studies have developed novel resilience indices and metrics to assess resilience and determine initiatives to enhance it. For example, Jin, Tang, Sun, et al. (2014) introduced localized integration with bus services to achieve a desired resilience to potential disruptions. They defined metro network resilience as the fraction of travel demand that can be satisfied by a degraded metro network after a disruption. They developed a mathematical model to generate a bus network layout for system resilience optimization. Lu (2018) explored a resilience approach for rail transit networks during daily operational incidents. By integrating the network topological and passenger volume characteristics, the approach explicitly considers the impacts of accumulatively affected passengers, quantifying the varying resilience of rail networks with time under different incidents. Vodopivec and Miller-Hooks (2019) quantified the resilience of transit systems from the perspective of user resilience and experience. They developed a multi-valued dependency graph framework that focused on diverse passenger populations. Shahabi, Raissi, Khalili-Damghani, et al. (2022) presented a simulation–optimization model that can generate resilient solutions for train scheduling problems under stochastic traveler flows, dynamic dwelling time, and random segment running times.

2.3. Focus of this study

In summary, the resilience assessment and optimization of urban rail systems is a very popular research topic, and many mathematical models and solution methodologies have been explored to respond to various risks. Preliminary studies have been conducted on the resilience of rail transit networks, which significantly contribute to managing disruptions and maintaining orderly metro operations. Nevertheless, two main research gaps can be identified: (i) Although many studies have been devoted to disruption management in urban rail systems, few studies have focused on the optimization of system resilience to short-term disruptions in daily operations (e.g., Gao et al., 2016; Huang et al., 2020; Kroon et al., 2015); (ii) existing studies on the resilience quantification of urban rail systems frequently ignore the effects of overcrowded passengers on the operations of trains at platforms. The passenger demands in these studies are usually considered as invariant values with unlimited train carrying capacity (e.g., Chen, Liu, Du et al., 2022; Chen, Liu, Peng et al., 2022; Jin et al., 2014; Lu, 2018; Yin et al., 2023b). Table 1 lists the detailed characteristics of some closely related works to show the contributions of our study, which are detailed as follows.

1. Our paper proposes an MINLP model for optimal train timetables to improve the resilience of an urban rail system. With system resilience as the overall optimization objective (substantially different with the objective function in Yin et al., 2023b), the model not only balances the train delays, service cancellations, and extra cost of reserve trains, but also maximizes the use of train capacity to relieve station congestion during peak hours. To solve the MINLP model, we develop a two-phase iterative optimization approach, where phase 1 focuses on the adjustment

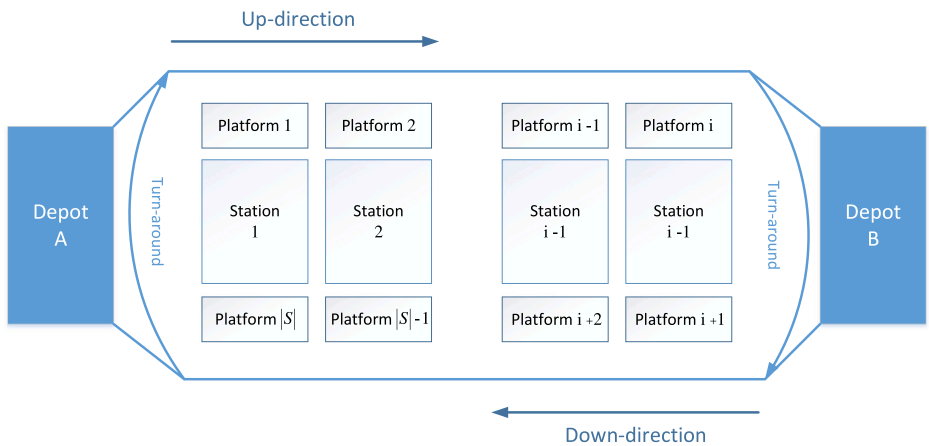


Fig. 1. A bi-directional urban rail line with two depots.

of trains to enhance the system resilience, and phase 2 evaluates the impact of passengers on the adjusted train schedule.

2. A resilience metric is defined as the congestion level of the busiest station during operation time. This metric is used as our objective function to minimize the passenger crowdedness at stations in case of disruptions. The metric explicitly captures different loading capacities of involved stations; thus, it is more suitable for assessing system resilience in peak hours than the common network topology metrics.
3. Constraints of the minimum dwelling time are considered, which can be effectively used for over-crowded conditions. In this paper, the train dwelling time is defined as a nonlinear function of the number of passengers boarding and alighting and the passenger congestion caused by overcrowding. The constraints are valid for train scheduling decisions during peak hours at overcrowded platforms, but trains dwell for the given time during off-peak hours.
4. Two sets of complex instances are investigated, involving an illustrative example and a real-world case based on the rolling stock and passenger flow data of Beijing Metro Line 1. We compare the computational results obtained using our approach with a commercial MILP solver and a heuristic method. The experiments demonstrated that our approach can evidently reduce the computational time compared with CPLEX. Moreover, the system resilience can be improved by as much as 39.7% using our optimization approach.

3. Problem statements

In this work, we consider a bi-directional urban rail transit line with a set $\mathcal{S}$ of stations, as shown in Fig. 1. In order to distinguish the operation direction of trains, we denote stations in the up-direction by 1, 2, …, i and stations in the down-direction by $i+1$, $i+2$, …, $|\mathcal{S}|$. In the up-direction, the start terminal, indexed by station 1, is connected to the returning terminal, indexed by station $|\mathcal{S}|$, through the corresponding depot $d_A \in \mathcal{S}^D$. Similarly, station $i+1$ is connected to station s_i through the other depot $d_B \in \mathcal{S}^D$.

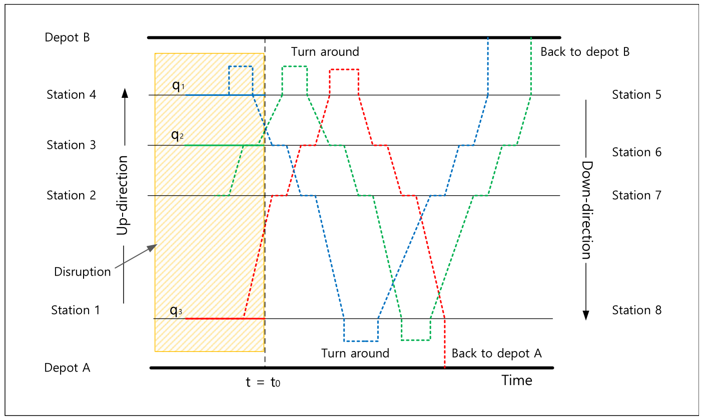
Given a set $\mathcal{Q}$ of train services, the rail managers make a train timetable and rolling stock circulation plan, that specifies the planned arrival time $\bar{t}^a_{qi}$ or departure time $\bar{t}^d_{qi}$ for each train $q \in \mathcal{Q}$ at stations $i \in \mathcal{S}$. An illustrative train schedule with trains q_1, q_2 and q_3 is presented in Fig. 2(a). According to the predetermined timetable, a train begins its journey at station 1 in the up-direction until it reaches station 4. After the given turn-around time at the returning terminal, it goes back to station 8 (i.e., station 1 physically) and then waits for the next cycle or goes back to the depot. In reality, a disruption may delay or cancel a series of train services. When the disruption is removed, rail managers make a conflict-free rescheduled timetable and a new rolling stock circulation plan using rolling stocks on the line or from the depots, in order to recover to the planned timetable as soon as possible. As shown in Fig. 2(b), if we take no action in the recovery of disruptions, the conflicts may happen at the turn-around station 1, leading to knock-on delays of the following trains.

In practice, rail dispatchers adopt several adjustment strategies according to specific operational scenarios to prevent delays and mitigate passenger stranding. These strategies typically involve suspending the movement of trains (i.e., holding all trains until the disruption is over), canceling some services, short-turning at key stations, and adjusting the rolling stock circulation plan. For instance, Fig. 3(a) presents a rescheduling plan in which three trains are suspended to avoid conflicts at the depot, which noticeably increases the delay time of the trains. Fig. 3(b) shows an example in which the trains run on time, but some of the services are canceled and the rolling stock circulation plan is adjusted. Fig. 3(c) further presents a combined rescheduling plan, in which train q_2 turns its direction at station 3 to cover the service initially served by train q_1. The train rescheduling plan shown in Fig. 3(c) not only covers all train services but also maintains a short delay time, thus yielding the best solution. Generally, train adjustment is combined with a set of strategies, considering the degree of delay, passenger flow characteristics, line conditions, and other factors. It uses the complementary advantages of different strategies to achieve good adjustment.

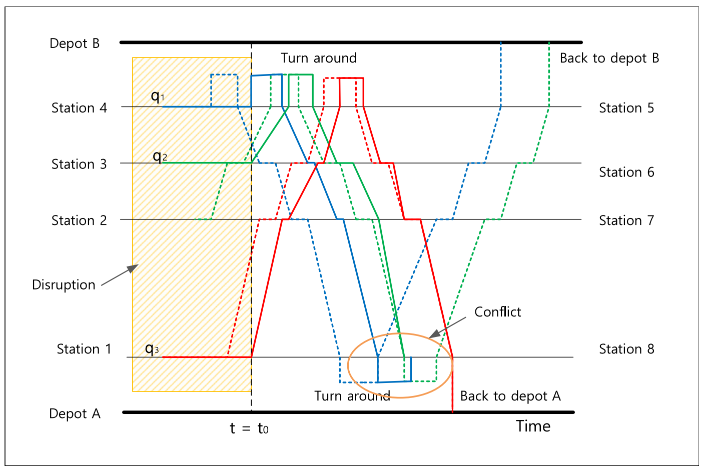
3.1. Definition of system resilience

With the increase in customer-oriented service quality management, managers have realized the importance of evaluating passenger flow status and service quality from the perspective of passengers. Rail operators are increasingly concerned about variations in passenger flow and travel demand that can be satisfied by degraded urban rail networks after disruptions. From the perspective of services to passengers, the resilience of an urban rail transit network, denoted by $\mathcal{R}$, can be mathematically expressed as the *congestion of the busiest station* over time (i.e., the station with the highest congestion level among all stations in $\mathcal{S}$):

$$\mathcal{R} = \max_i \frac{\sum_{\tau \in \mathcal{T}_{disrupt}} PW_i^\tau}{NT \cdot SC_i}, \quad \forall i \in \mathcal{S}, \ \mathcal{T}_{disrupt} = \{t \in \mathcal{T} \mid t_{start} \le t \le t_{end}\} \tag{1}$$



(a) Planned timetable of three trains q_1, q_2 and q_3



(b) Rescheduled timetable with conflict of trains

Fig. 2. Planned timetable (a) and rescheduled timetable with conflict (b).

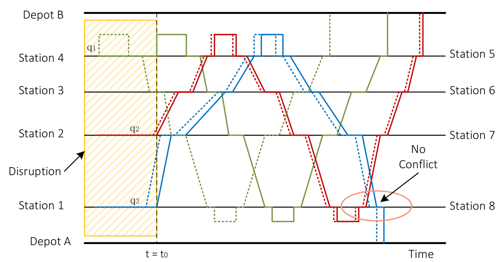
where $\mathcal{T}$ represents the entire time cycle (e.g., 6 am to 11 pm); t_{start} and t_{end} represent the beginning and end times of the considered period (near a disruption), respectively; $NT = t_{end} - t_{start}$ indicates the length of the considered period; PW_i^{τ} is the number of waiting passengers at station i at time τ; and SC_i indicates the maximum number of passengers the station i can accommodate. The value of SC_i is predetermined by the layout of each station (in the construction phase of an urban rail network). A platform has two types of passengers waiting during the departure of two adjacent trains. One type is passengers entering the platform from outside the station during this period. The other type includes stranded passengers who failed to board previous trains because of platform congestion. Thus, the system resilience indicator $\mathcal{R}$ represents the time-varying congestion of the densest station during a disruption. In particular, a smaller value of $\mathcal{R}$ indicates better resilience to disruptions because congestion at crowded stations can be eliminated more quickly.

We may argue that the resilience of urban rail systems is measured by changes in aggregated travel and waiting times, as they are also proxies for disruption costs. However, passengers are affected by psychological pressures and personal emotions in crowded environments, and their perception of time also changes (Droit-Volet & Meck, 2007). Rose, Yuan, and Zhang (2007) showed that queuing or waiting in a bad environment can increase passenger perception time. In addition, passengers' negative emotions due to crowded environments not only affect their perception of the service level but also station transport safety (Zhang & Li, 2017). Therefore, more attention should be given to station management of passenger flows. With the aim of being "passenger-oriented", the metro operators gradually provide a "proactive" service that optimizes the facility configuration and layout design and strives to enable a better travel experience for passengers (Teng, Pan, & Zhang, 2020).

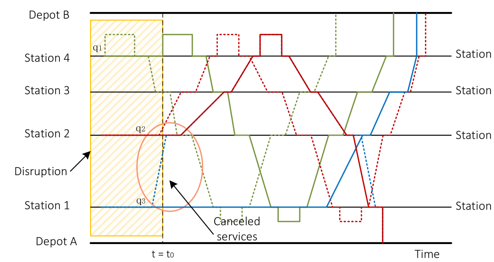
3.2. Rescheduling of trains based on an event-activity network

The services provided by urban rail transit can be viewed as the movement of trains between stations. In this paper, we denote an event-activity network $\mathcal{N} = (\mathcal{E}, \mathcal{P})$, which involves a set of events $e \in \mathcal{E}$ to represent the arrival and departure of trains at each station and several sets of activities $p \in \mathcal{P}$ to represent the relationships between two events. In our formulation, each event $e \in \mathcal{E}$ indicates the movement of trains operating between two successive stations. Event e in network $\mathcal{N}$ is associated with the departure time t_e^d from the first station s_e^o and arrival time t_e^a at the next station s_e^d. In addition, each activity p, defined as $(\underline{e}, \overline{e})$, is a directed arc from one event (i.e., $\underline{e}$) to another (i.e., $\overline{e}$). A train line uses an orderly connection between events e to provide passenger mobility services. In graph $\mathcal{N}$, we define three sets of activities: running, turnaround, and depot-related activities, detailed as follows:

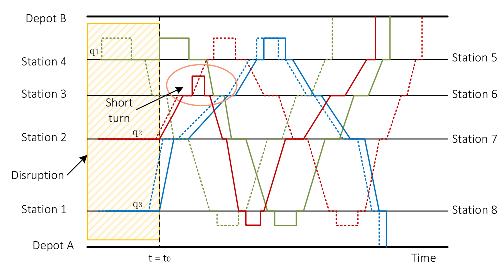
- The running activity $p \in \mathcal{P}$ refers to two events $\underline{e}$ and $\overline{e}$ in the same running direction, which satisfy $s^d_{\underline{e}} = s^o_{\overline{e}}$. In other words, the running activity means that a train arrives at a station in one direction and continues to the next station in the same direction.
- The turnaround activity p is defined from event $\underline{e}$ to event $\overline{e}$ that operates in the opposite direction. Turnaround activities can be divided into two different types: the event $\underline{e}$ short turning to event $\overline{e}$ at the intermediate station with the short-turning



(a) Rescheduled timetable by suspending strategy



(b) Rescheduled timetable by cancelation of services



(c) Rescheduled timetable by combined strategies

Fig. 3. Rescheduled timetable with different strategies (a. Suspending; b. Service cancellation; c. Combined).

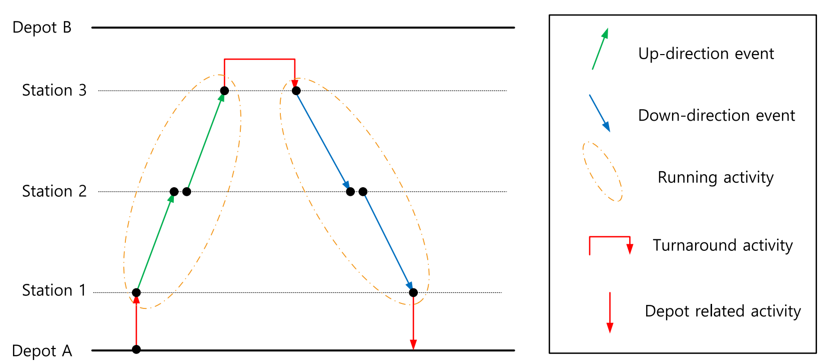


Fig. 4. Event-activity network representation.

capacity, and event e turning around at the terminal station for the train circulations.

- The depot-related activity p is defined as an event e that originates from the depot or returns to the depot. This represents the start or the end of the event in the event-activity network.

An example with three stations and two depots is shown in Fig. 4. The events in this figure form a whole space–time trajectory of a train, which originates from depot A, travels in the upward direction to station 3, turns its direction, and finally returns to depot A. We can observe from the example that the event network can represent planned and rescheduled train timetables by selecting a set of connected events. For instance, we can let the train turn around at station 2 (i.e., short-turning operations) to create a new train trajectory. In the following section, we describe the construction of a mathematical model to optimize rescheduled trains in an event-activity network.

3.3. Time-dependent passenger flows

To describe the travel demand of passengers, a feasible traveling path pa is defined if passengers can move from one station i to another j without any repetition of stations, i.e. completing the travel plan from station to station. In a travel direction, pa can be denoted by (i, j) where $i \in \mathcal{S}$ and $j \in \mathcal{S}$ respectively denote the origin and destination of path pa. Such a series of paths pa constitute a set $\mathcal{P}$ of feasible traveling paths, and we also define two subsets $\mathcal{P}_i^o \subset \mathcal{P}$ and $\mathcal{P}_j^d \subset \mathcal{P}$ that respectively represent the set of traveling paths with origin i and destination j, where $i, j \in \mathcal{S}$.

In the urban rail transit systems, it is widely recognized that the passenger demand is time-dependent and the passenger arriving rates vary at different time periods. For the single-line train scheduling, the dynamic passenger demand is captured by using OD matrices, in which each term (i, j) is a function of time that indicates the number of passengers willing to travel from station $i \in \mathcal{S}$ to station $j \in \mathcal{S}$. Each element in this matrix is thus time dependent, and the passenger arrival time thus depends on the timetable and chosen paths. Due to this reason, our study employs a path-based representation to depict the time-dependent passenger demand in an urban rail transit network. Specifically, we denote a time-dependent matrix $\mathcal{OD}_t = \{\mathcal{OD}_{pa}^t | pa \in \mathcal{PA}, t \in \mathcal{T}\}$, where each element $\mathcal{OD}_{pa}^t$ represents the number of passengers with path pa entering the network at time t. Thus, the number of passengers who originate from station i at time t is $PE_i^t = \sum_{pa \in \mathcal{PA}_i^o} \mathcal{OD}_{pa}^t$, i.e. the arrival rate of passengers in station i at time t.

3.4. Problem definition and assumptions

Considering a busy urban rail line, the crowdedness of a station on the rail line is directly influenced by four elements: the time-dependent passenger origin–destination (OD) matrix, timetables of the involved lines, train-carrying capacity, and station capacity threshold. As the train and station capacities are frequently fixed at the network planning level, the only method to alleviate the crowdedness of stations is to adjust the timetables. In this paper, the disruption management of urban rail transit systems focuses precisely on the real-time rescheduling of rolling stocks and train timetables in the recovery phase of disruptions or under a known disruption duration. However, rail operators tend to consider only the train operating indicators when rescheduling train timetables. To compensate for the delay in train arrival promptly, the dwelling time of trains is compressed such that at certain stations, the dwelling time cannot satisfy the demand of passengers for boarding and alighting. Particularly during peak hours, owing to the significantly increased travel demand, many passengers may be stranded at stations affected by disruptions.

Therefore, the aim of our paper is to provide a unified mathematical formulation for the disruption management of urban rail transit systems as well as applying the passenger-oriented adjustment approach to enhance the resilience of urban rail systems. Specifically, it involves rescheduling train timetables by varying operation routes and dwelling time with consideration of adjustment strategies to minimize the negative impact on passenger travel caused by the reduction in the operation capacity of rail lines during the disruption. The mathematical formulation is based on the following five assumptions:

Assumption 1. There are no delays and the train does not short-turn in the opposite direction under normal conditions. Moreover, the running time of trains in segments is prespecified according to field data. Trains running in the upward and downward directions are independent.

Assumption 2. Trains depart from the start terminal in a first-in-first-out manner. Each station can accommodate only one train at a time, and overtaking and crossing operations are prohibited at any position, which is consistent with the practical requirements of urban railway systems.

Assumption 3. Time-dependent passenger demand in the railway network is denoted by the number of passengers on each traveling path, which fixes the OD station and departure time from the origin station. The traveling paths of passengers are prespecified by historical data, and thus we do not consider the passengers' route choice. This assumption is valid here because our focus is on managing traffic flows during peak hours, when the train frequency is significantly high, similar to the research in Chai, Yin, D'Ariano, Samà, and Tang (2023a).

Assumption 4. Passengers are considered more likely to select trains that require less waiting time and do not leave the system owing to platform congestion. We assume that all passengers will alight from the train at the last station, and no passengers will board because there is no next station at the terminal, similar to the research in Bucak and Demirel (2022).

Assumption 5. Disruption occurrence is known at a time t_{start}, before the actual beginning of disruption and disruption handling cannot happen prior to the disruption location and time of occurrence in either directions of travel, similar to the research in Narayanaswami and Rangaraj (2013). It eliminates the uncertainty factor in defining disruption occurrence, so that disruption can be clearly incorporated in the optimal rescheduling model.

4. Mathematical formulations

In this section, using the resilience-oriented quantitative evaluation index as the objective function, we construct an MINLP model. A series of operating constraints for basic operation, short-turning routes, and rolling stock circulations with backup trains are formulated to accomplish the train rescheduling model. Finally, we model passenger flows that vary over time considering the demands of dynamic passengers and train capacity constraints. Finally, the constraint of minimum dwelling time is introduced into the initial formulation, considering the impact of passenger boarding and alighting on train delays.

4.1. Notations

For the convenience of readers, the key notations used throughout the paper are summarized in Table 2 and the corresponding decision variables are listed in Table 3.

4.2. Objective function

The aim of our model is to qualitatively evaluate and optimize the system resilience of an urban rail line to disruptions, i.e., to reduce the congestion level of stations. Thus, we define the objective function as a resilience-oriented evaluation index as follows:

$$\min \ (\max_i \ \frac{\sum_{\tau \in \mathcal{T}_{disrupt}} PW_i^\tau}{NT \cdot SC_i}), \ \ \forall i \in \mathcal{S}, \ \mathcal{T}_{disrupt} = \{t \in \mathcal{T} \mid t_{start} \le t \le t_{end}\} \tag{2}$$

4.3. Train rescheduling constraints

Based on the event-activity network, the constraints for basic operation, short-turning, and rolling stock rescheduling are introduced as follows:

(1) Constraints for basic train running: First, we consider the constraints to limit the arrival/departure times t_e^a and t_e^d of trains in a train rescheduling plan. Let T_e^R denote the minimum traveling time of event e (i.e., the minimum traveling time from station s_e^o to the successive station s_e^d). The traveling time in this section can be calculated using the train velocity curves. We derive the following constraints for the arrival and departure times of trains:

$$t_e^a - t_e^d \ge T_e^R, \ \ \forall e \in \mathcal{E} \tag{3}$$

$$t_e^d - \bar{t}_e^d \ge 0, \ \ \forall e \in \mathcal{E}: \ \bar{t}_e^d \le t_{start} \tag{4}$$

$$c_e \le 0, \ \ \forall e \in \mathcal{E}: \ \bar{t}_e^d \le t_{start} \tag{5}$$

$$t_e^d \ge (1 - c_e) t_{end}, \ \ \forall e \in \mathcal{E}^D \tag{6}$$

Constraint (3) indicates that each event e ensures minimal operating time. The travel time between stations s_e^o and s_e^d should not be less than the minimum running time, T_e^R. According to (4), the departure time t_e^d for event $e \in \mathcal{E}$ cannot be earlier than the original time $\bar{t}_e^d$. Considering the impact of the disruption, we assume that the service events before the disruption starting time t_{start} cannot be canceled, as constraint (5) shows. Constraint (6) indicates that in the disruption-lasting phase, the service event e that originally operates in the disruption area should be canceled or wait until the end of the disruption.

For event connection p, event e affects the operation of event $\bar{e}$; the relationship between events e and $\bar{e}$ is formulated as

$$t_{\bar{e}}^d - t_e^a + (c_e + c_{\bar{e}}) * M \ge T_{e,\bar{e}}^D, \ \ \forall e \in \mathcal{E}: \ (e, \bar{e}) \in \mathcal{P} \tag{7}$$

$$(c_e + c_{e'}) * M + t_{e'}^d \ge t_e^d + T_{e,e'}^H, \ \ \forall e \in \mathcal{E}, \ e' \in H^+(e) \tag{8}$$

Constraint (7) guarantees that the train must dwell at each station for at least the required dwelling time $T_{e,\bar{e}}^D$ to satisfy passenger demands. M is a very large positive value Therefore, if $c_e = 1$ or $c_{\bar{e}} = 1$ (i.e., event e or $\bar{e}$

Table 2
Parameters and notations for the model formulation.

Notation	Detailed definition
$\mathcal{S} = \{1, 2, \dots, i, i+1, \dots, \lvert\mathcal{S}\rvert\}$	The set of stations along the line.
$\mathcal{S}^{\mathcal{D}}$	The set of depots in the network.
$\mathcal{Q} = \{1, 2, \dots, \lvert\mathcal{Q}\rvert\}$	The set of train services of the line.
$\mathcal{T} = \{0, 1, \dots, \lvert\mathcal{T}\rvert\}$	The set of time periods.
$\mathcal{E}$	The event network indexed by e.
$\mathcal{P}$	The event pair in the network $\mathcal{E}$, and the element is denoted by $(\underline{e}, \overline{e})$, where $\underline{e}$ means the first event in the pair and $\overline{e}$ represents the following event in the network.
$\mathcal{E}^{\mathcal{D}}$	The event set that influenced by the disruption.
$\mathcal{E}^{\mathcal{S}} = \mathcal{E}^{\mathcal{S}^+} \cup \mathcal{E}^{\mathcal{S}^-}$	The possible short-turning event set, where $\mathcal{E}^{\mathcal{S}^+}$ means the event set in which the event e can short turn to another event e' and $\mathcal{E}^{\mathcal{S}^-}$ represents the event set in which the event e can be short turned by another event e'.
$\mathcal{E}^{T} = \mathcal{E}^{T^+} \cup \mathcal{E}^{T^-}$	The event set that needs to turn around at the terminal station with a depot, where $e \in \mathcal{E}^{T^+}$ means the event that comes out from the terminal station and $e \in \mathcal{E}^{T^-}$ means the event that goes into the station.
s_e^a	The arriving station of event e.
s_e^d	The departure station of event e.
$\mathcal{D}^+(e)$	The event set that can be connected by event e (i.e., e turns around to $e' \in \mathcal{D}^+(e)$).
$\mathcal{D}^-(e)$	The event set that can connect to event e (i.e., $e' \in \mathcal{D}^-(e)$ turns around to e).
$H^+(e)$	The following event set of event e in the same section.
$\bar{t}_e^d$	The departure time of event e in the planned timetable.
$\bar{t}_e^a$	The arrival time of event e in the planned timetable.
T_e^R	The running time of service e between two stations s_e^o and s_e^d.
T_e^D	The minimum dwelling time at station s_e^o.
$T_{e,e'}^H$	The minimum departure headway time between event e and e'.
$T_{e,e'}^T$	The minimum turnaround time between event e and e'.
$T_{e,e'}^P$	The minimum time distance between event e and e' where e turns to event e'.
T_{depot}	The minimum time that a train comes back to the depot and departs from the depot for a new trip.
$[t_{start}, t_{end}]$	The duration of the disruption. t_{start} is the start time and t_{end} is the end time of the disruption.
N_{train}	The total number of trains in the network at time t_{start}.
N_{add}^d	The number of reserved physical trains at depot d, $d \in \mathcal{S}^{\mathcal{D}}$.
$\mathcal{PA} = \{pa = (i,j) \mid i \neq j, i \in \mathcal{S}, j \in \mathcal{S}\}$	The set of feasible traveling paths of passengers, and the element represents a specific traveling path, denoted by pa.
$\mathcal{PA}_i^o$	The set of traveling paths with origin i, where $i \in \mathcal{S}$.
$\mathcal{PA}_i^d$	The set of traveling paths with destination i, where $i \in \mathcal{S}$.
$\mathcal{PA}_{i',i}$	The set of paths that originate from station i' and end at stations located after station i.
$\mathcal{OD}_{pa}^t$	The number of passengers with path pa entering the network at time t.
t_i^r	The running time of the trains between the connected stations i and $i+1$.
t_i^d	The dwelling time of trains at station i.
$\bar{t}_i$	The running time of the first train from the depot to station i.
$\mathcal{T}_t = \{t \in \mathcal{T} \mid t' \leq t\}$	The set of the discrete time units $t' \in \mathcal{T}$ that are not later than time t.
$\mathcal{T}_{pa,i} = \{t \in \mathcal{T} \mid t \geq t_i^{pa}\}$	The set of the discrete time units $t \in \mathcal{T}$ that are larger than t_i^{pa} on path pa.
T_{min}	The minimum headway time of trains.
T_{cycle}	The minimum circulation time of rolling stock.
n_{pa}	The total number of passengers with path $pa \in \mathcal{PA}$.
C	The train capacity.
K	The number of train services that can depart from the depot within each time window $[t, t+T_{cycle}]$ at most.
N_{door}	The total number of doors on each train.
$\mathcal{R}$	Resilience metric
w_1	The weight-coefficient that represents the costs of train delays
w_2	The weight-coefficient that represents the costs of canceled services
w_3	The weight-coefficient that represents the costs of back-up rolling stock

Table 3
Decision variables in models.

Decision variable	Description	Variable type
t_e^d	The rescheduled departure time of event e.	R^+
t_e^a	The rescheduled arrival time of event e.	R^+
c_e	If event e is canceled, then $c_e = 1$; otherwise $c_e = 0$.	$\{0,1\}$
$\lambda_{e,e'}$	If event e turns to e', then $\lambda_{e,e'} = 1$; otherwise $\lambda_{e,e'} = 0$.	$\{0,1\}$
α_e	If event e is served by a train from the depot, then $\alpha_e = 1$; otherwise $\alpha_e = 0$.	$\{0,1\}$
β_e	If the train serving event e goes back to the depot, then $\beta_e = 1$; otherwise $\beta_e = 0$.	$\{0,1\}$
x_t	If a train departs from the line at time $t \in \mathcal{T}$, then $x_t = 1$; otherwise $x_t = 0$.	$\{0,1\}$
PW_i^t	The number of waiting passengers at station $i \in \mathcal{S}$ and time $t \in \mathcal{T}$.	Z^+
$PB_{pa,i}^t$	The number of boarding passengers with traveling path $pa \in \mathcal{PA}$ at station $i \in \mathcal{S}$ and time $t \in \mathcal{T}$.	Z^+
PE_i^t	The number of passengers who enter from outside the station $i \in \mathcal{S}$ at time $t \in \mathcal{T}$.	Z^+
TS_d	The crowding level in the train of each door	R^+
$t(i', i)$	The passengers' boarding time at the former stations $i' \in \mathcal{S}_i$.	R^+
$t_{i',i}$	The traveling time from station $i' \in \mathcal{S}_i$ to $i \in \mathcal{S}$.	R^+

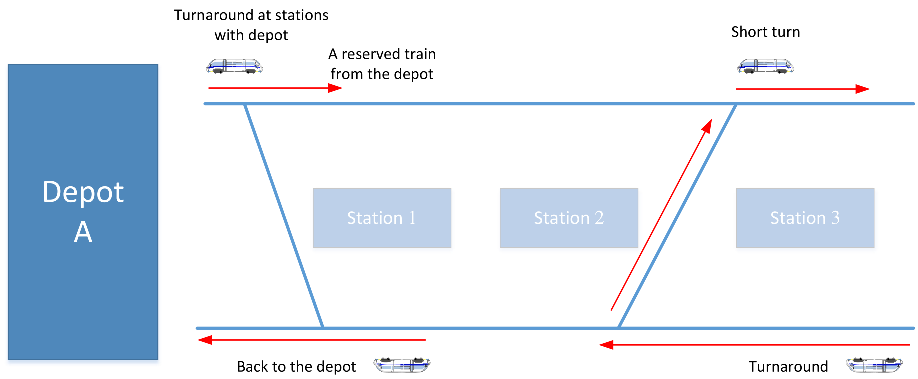


Fig. 5. Turnaround at terminal stations with reserve trains.

is canceled), inequality (7) is always true; otherwise, arrival time t_e^a and departure time t_e^d at every station must satisfy the minimum dwelling-time requirement. To ensure safe operation, a train must travel a safe distance. In constraint (8), $H^+(e)$ is the following event set in this section, and the minimal headway time $T^H_{e,e'}$ should be greater than the minimal operating limitation. In our model, a train cannot overtake at the station. This implies that the train should enter the station after the departure of the last train.

(2) Constraints for the train turnaround: The turnaround operations support trains to switch from the up–down direction to the down–up direction. In actual operation, the train is permitted to not only turn around at terminal stations but also use a short-turning strategy to turn around at non-terminal stations. Notably, certain terminal stations connected to the rolling stock depot can provide reserve trains for timetable rescheduling, as shown in Fig. 5. Hence, we derive the following two categories of sets:

$$\mathcal{E}^S = \mathcal{E}^{S^+} \cup \mathcal{E}^{S^-}$$
$$\mathcal{E}^T = \mathcal{E}^{T^+} \cup \mathcal{E}^{T^-}$$

where $\mathcal{E}^S$ and $\mathcal{E}^T$ represent the turnaround event sets that are not connected and connected to a depot, respectively. Specifically, $\mathcal{E}^{S^+}$ represents the event set in which event e can rapidly turn to another event e', $\mathcal{E}^{S^-}$ represents the event set in which event e can be quickly turned by another event e', $\mathcal{E}^{T^+}$ represents the event set that approaches the depot, and $\mathcal{E}^{T^-}$ is the event set that moves away from the depot.

For simplicity, we define the following sets with respect to each event $e \in \mathcal{E}^S \cup \mathcal{E}^T$:

$$\mathcal{D}^+(e) = \{e' \in \mathcal{E}^{S^-} \cup \mathcal{E}^{T^-} \mid s_e^d = s_{e'}^o\}$$

for $e \in \mathcal{E}^{S^+} \cup \mathcal{E}^{T^+}$, and

$$\mathcal{D}^-(e) = \{e' \in \mathcal{E}^{S^+} \cup \mathcal{E}^{T^+} \mid s_e^o = s_{e'}^d\}$$

for $e \in \mathcal{E}^{S^-} \cup \mathcal{E}^{T^-}$. Here, $\mathcal{D}^+(e)$ and $\mathcal{D}^-(e)$ represent the sets of events that can possibly connect to or be connected by event e at the same station.

Event e is performed by the rolling stock; thus, one event e can only turn around or can be turned around by event e' no more than once. If events e and e' successfully form a connection for turnaround operations, the minimum turnaround time $T^T_{e,e'}$ must be ensured.

$$\sum_{e' \in \mathcal{D}^+(e)} \lambda_{e,e'} \le 1, \quad \forall e \in \mathcal{E}^{S^+} \cup \mathcal{E}^{T^+} \tag{9}$$

$$\sum_{e' \in \mathcal{D}^-(e)} \lambda_{e',e} \le 1, \quad \forall e \in \mathcal{E}^{S^-} \cup \mathcal{E}^{T^-} \tag{10}$$

$$(1-\lambda_{e,e'}) * M + t_{e'}^d \ge t_e^a + T^T_{e,e'}, \ \forall e \in \mathcal{E}^{S^+} \cup \mathcal{E}^{T^+}, \ e' \in \mathcal{D}^+(e) \tag{11}$$

Finally, we consider the case of short-turning operations. Note that if two events e and e' are connected for short-turning operations, we ensure that neither e nor e' can be canceled. Moreover, any event $\overline{e}$ that satisfies $(e,\overline{e}) \in \mathcal{P}$ must be canceled because $\overline{e}$ can no longer be covered by any train. Accordingly, we obtain the following two sets of constraints:

$$\sum_{e' \in \mathcal{D}^+(e)} \lambda_{e,e'} + c_e \le c_{\overline{e}}, \quad \forall e \in \mathcal{E}^{S^+} : (e,\overline{e}) \in \mathcal{P} \tag{12}$$

$$\sum_{e' \in \mathcal{D}^-(e)} \lambda_{e',e} + c_e \le c_{\underline{e}}, \quad \forall e \in \mathcal{E}^{S^-} : (\underline{e},e) \in \mathcal{P} \tag{13}$$

Turnaround tracks at non-terminal stations can accommodate at most one train to turn its direction, as shown in Fig. 5. Otherwise, a train conflict may occur. Therefore, the following constraints are derived to avoid these scenarios:

$$(1-\lambda_{e,e'} + c_{e''}) * M + t_{e''}^a \ge t_{e'}^d + T^P_{e',e''}, \quad \forall e \in \mathcal{E}^{S^+}, e' \in \mathcal{D}^+(e), e'' \in H^+(e) \tag{14}$$

$$(1-\lambda_{e,e'} + c_{e''}) * M + t_{e''}^d \ge t_e^a + T^P_{e,e''}, \quad \forall e \in \mathcal{E}^{S^+}, e' \in \mathcal{D}^+(e), e'' \in H^+(e') \tag{15}$$

Here, $T^P_{e',e''}$ denotes the minimum tracing time between events e' and e'' due to the occupancy of the turn-out track. Constraints (14) and (15) guarantee minimal upstream and downstream tracing times, respectively.

Note that these constraints apply only to turnaround operations for non-terminal stations. In the next section, turnaround operations for terminal stations connected to depots are further considered. Turnaround operations with depots can be more complex because the reserve trains in the depot can be used for timetable rescheduling to speed up recovery from disruption.

(3) Rolling stock rescheduling at terminal stations with depots: In practice, the dispatcher may not cancel the service event to improve the service level. Because of this disruption, the service event delays compared with the original plan. The rolling stock from the depot that conducts the service event can contribute to reducing the impact of the disruption. To model rolling stock circulation at depots with reserve trains, we introduce two groups of binary decision variables, α_e and β_e:

$$\alpha_e = \begin{cases} 1 & \text{Event } e \text{ is served by a train from the depot} \\ 0 & \text{otherwise} \end{cases}$$

$$\beta_e = \begin{cases} 1 & \text{The train serving event } e \text{ goes back to the depot} \\ 0 & \text{otherwise} \end{cases}$$

for any $e \in \mathcal{E}^T$. Event e turning around with the depot should carry a single rolling stock. Subsequently, the following constraints are derived:

$$\sum_{e' \in \mathcal{D}^+(e)} \lambda_{e,e'} + \beta_e \le 1, \quad \forall e \in \mathcal{E}^{T^+} \tag{16}$$

$$\sum_{e' \in \mathcal{D}^-(e)} \lambda_{e',e} + \alpha_e \leq 1, \quad \forall e \in \mathcal{E}^{T^-} \tag{17}$$

According to (16), if the train returns to the depot, event e carried by this train cannot connect to another event e'. Constraint (17) indicates that event e can be performed by only one train.

The relationship between the event connection $\lambda_{e,e'}$ and canceled event e is formulated as follows:

$$\alpha_e + \sum_{e' \in \mathcal{D}^-(e)} \lambda_{e',e} \geq 1 - c_e, \quad \forall e \in \mathcal{E}^{T^-} \tag{18}$$

$$\beta_e + \sum_{e' \in \mathcal{D}^+(e)} \lambda_{e,e'} \geq 1 - c_e, \quad \forall e \in \mathcal{E}^{T^+} \tag{19}$$

$$c_e + c_{e'} \leq 2 * (1 - \lambda_{e,e'}), \quad \forall e \in \mathcal{E}^{T^+}, \ e' \in \mathcal{D}^+(e) \tag{20}$$

Constraint (18) indicates that service event e is performed by a train that must travel from the depot or connect to another service event e'; otherwise, service event e must be canceled. Similarly, constraint (19) limits the turnaround events in which the train may return to the depot. Service event e carried by a physical train should turn to another service event e' or cancel service event e at the terminal station. Finally, if service events e and e' are directly connected and served by the same rolling stock, the dispatcher cannot cancel these two service events. Hence, constraint (20) should be satisfied, guaranteeing that $c_e = 0$ and $c_{e'} = 0$ if $\lambda_{e,e'} = 1$.

Train resources are limited in actual operation; N_{train} denotes the total number of available trains operating on the line. To enhance the operational resilience, N_{add} represents the number of reserve trains. When a service event e requires a train, rolling stock should be available for operation. The constraints are described as follows:

$$\sum_{e' \in \mathcal{E}^{T^+} : t^d_{e'} < t^d_e} \alpha_{e'} - \sum_{e'' \in \mathcal{E}^{T^-} : t^a_{e''} + T_{depot} < t^d_e} \beta_{e''} \leq N_{train} + \sum_{d \in D} N^d_{add}, \quad \forall e \in \mathcal{E}^{T^+} \tag{21}$$

where the first part is the total number of trains that leave the depot, and the second part indicates the available number of trains that return to the depot and can be used to perform service event e. The overall inequality guarantees that available trains operate on the line should be smaller than the maximum fleet size plus the reserve rolling stock at the depot.

Nevertheless, constraint (21) has a nonlinear summation over the decision variables α_e and β_e. In the following section, we translate this into a linear format. We introduce another binary decision value $\gamma_{e,e'}$, which implies that the train of service event e' has sufficient time to recirculate to service event e. $\gamma_{e,e'}$ is defined as

$$\gamma_{e,e'} = \begin{cases} \beta_{e'} & \text{if } t^a_{e'} + T_{depot} \leq t^d_e \\ 0 & \text{else } t^a_{e'} + T_{depot} > t^d_e, \end{cases} \quad e \in \mathcal{E}^{T^+}, \ e' \in \mathcal{E}^{T^-}$$

In our model, the variable $\beta_{e'}$ is mapped to $\gamma_{e,e'}$ to remove the nonlinear part of constraint (21). From the definition of $\gamma_{e,e'}$, if $\beta_{e'}$ is equal to 0, the variable is also equal to 0. Meanwhile, if the expression $t^a_{e'} + T_{depot} \geq t^d_e$ is true and $\beta_{e'}$ is equal to 1, $\gamma_{e,e'}$ is equal to 0. This logical expression can be represented by the following $if - then$ constraint (22).

$$\gamma_{e,e'} = 0 \ \Rightarrow \ t^a_{e'} + (1 - \beta_{e'}) * M + T_{depot} \geq t^d_e \tag{22}$$

The above $if - then$ constraint can be easily reformulated as a linear inequality by introducing a big-M value, given by

$$t^a_{e'} + (1 - \beta_{e'}) * M + T_{depot} - t^d_e \leq \hat{M} * (1 - \gamma_{e,e'}) \tag{23}$$

where $\hat{M} > M$. Finally, the nonlinear part of constraint (21) is replaced by $\gamma_{e,e'}$, and constraint (21) can be equivalently reformulated as follows:

$$\sum_{e' \in \mathcal{E}^{T^+} : t^d_{e'} < t^d_e} \alpha_{e'} - \sum_{e'' \in \mathcal{E}^{T^-}} \gamma_{e,e''} \leq N_{train} + \sum_{d \in D} N^d_{add}, \quad \forall e \in \mathcal{E}^{T^+} \tag{24}$$

4.4. Passenger flow constraints

After the system is disrupted, the original train schedules are adjusted, which also affects the travel of passengers. In particular, for high-density urban rail systems such as the Beijing Metro, passenger flow arrangements and outbursts have become a critical part of disruption management. Then we are going to model passenger flows on the line. In our model, the time horizon is discretized into a set of timestamps $\mathcal{T} = \{0, 1, \ldots, |\mathcal{T}|\}$. The running time of the train between the connected stations i and $i+1$ is defined as t^r_i, and the dwelling time at station $i \in \mathcal{S}$ is given by t^d_i. Given the above parameters (i.e. running time and dwelling time), we can actually obtain the total traveling time of passengers on each path pa from the origin station i, denoted by t^{pa}_i.

For modeling convenience, we adopt the following two sets, for each $0 \leq t \leq |\mathcal{T}|$:

$$\mathcal{T}_t = \{t' \in \mathcal{T} \mid t' \leq t\}$$

$$\mathcal{T}_{pa,i} = \{t \in \mathcal{T} \mid t \geq t^{pa}_i\}$$

The first set $\mathcal{T}_t$ defines the discrete time units $t' \in \mathcal{T}$ that are not later than time t. The second set $\mathcal{T}_{pa,i}$ is defined on path pa of each station i and includes the discrete time units $t \in \mathcal{T}$ that are larger than t^{pa}_i.

Firstly, we assume that all the passengers can board the next coming train, ignoring the train capacity. Based on this simplification, we define the following three sets of decision variables:

$$x_t = \begin{cases} 1 \text{ if a train departs from the origin station} \\ \quad \text{on the line at time } t \in \mathcal{T}; \\ 0 \text{ otherwise.} \end{cases}$$

$PW^t_i =$ the number of waiting passengers at station $i \in \mathcal{S}$ and time $t \in \mathcal{T}$.

$PB^t_{pa,i} =$ the number of boarding passengers with traveling path $pa \in \mathcal{PA}$ at station $i \in \mathcal{S}$ and time $t \in \mathcal{T}$.

In the above sets of variables, x_t determines the departure time of trains on the line. According to Assumption 1, this set of variables actually decides a coordinated train timetable (i.e. train arrival and departure times of each station). Variables PW^t_i and $PB^t_{pa,i}$ are associated with the movement of passenger flows on the line. Noted that the passengers are actually unable to board any train until the first train arrives into the station. For example, we assume that the first train from the depot takes $\bar{t}_i$ time units to arrive at station i. Then, $PB^t_{pa,i}$ keeps equal to zeros if $t \leq \bar{t}_i$.

$$PW^t_i = \sum_{\tau \in \mathcal{T}_t} PE^\tau_i - \sum_{\tau \in \mathcal{T}_t} \sum_{pa \in \mathcal{PA}^o_i} PB^\tau_{pa,i}, \quad \forall i \in \mathcal{S}, \ t \in \mathcal{T} \tag{25}$$

$$\sum_{pa \in \mathcal{PA}_i} PB^t_{pa,i} \leq C * x_{t-t_i}, \quad \forall i \in \mathcal{S}, \ t \in \mathcal{T} : t \geq t_i \tag{26}$$

$$\sum_{\tau \in \mathcal{T}_t} PB^\tau_{pa,i} \leq \sum_{\tau \in \mathcal{T}_t} \mathcal{OD}^\tau_{pa}, \quad \forall i \in \mathcal{S}, \ pa \in \mathcal{PA}^o_i, t \in \mathcal{T}_{pa,i} \tag{27}$$

Constraint (25) fixes the flows of waiting passengers, arriving passengers and boarding passengers at each station $i \in \mathcal{S}$ during the considered time horizon $\mathcal{T}$. As it describes, the number of waiting passengers at station $i \in \mathcal{S}$ and time unit $t \in \mathcal{T}$ is calculated by considering the cumulative arriving passengers and by subtracting all the boarding passengers for time units $\{0, 1, \ldots, t\}$. Constraint (26) is the passenger boarding constraint, which guarantees that the passengers at station i are able to board when the next train arrives at station i with path pa. t_i is a pre-given parameter that denotes the traveling time of any train from the origin station to station i. In particular, if no train departs from the origin station at time $t - t_i$ (i.e. $x_{t-t_i} = 0$), the number of boarding passengers at station i and time t is zero. Besides, constraint (26) also imposes an upper bound C on the number of passengers that can board each train. Here, the passenger boarding constraint ignores the passengers that are already on the train, before starting the current boarding process. And the passenger boarding constraint will be extended later with the influence of passengers on the train. Constraint (27) sets the threshold for the number of boarding passengers at each time unit t, which should be no more than the maximum number of available passengers for each path pa. For the origin station i of path pa, constraint (27) ensures that the number of boarding passengers with origin station i cannot exceed the total number of arriving passengers until time t.

$$\sum_{t \le t' \le t+T_{min}} x_{t'} \le 1, \quad \forall t \in \mathcal{T} \tag{28}$$

$$\sum_{t \le t' \le t+T_{cycle}} x_{t'} \le K, \quad \forall t \in \mathcal{T} \tag{29}$$

Constraint (28) ensures that no more than one train can depart from the depot within the time window $[t, t+T_{min}]$ (i.e. the train headway constraint). And the rolling stock circulation constraint (29) guarantees that, given a set of K physical trains on the line, at most K train services can depart from the depot within each time window $[t, t+T_{cycle}]$.

The boarding passenger constraint (26) limits the number of boarding passengers with a fixed value, i.e., train capacity C. As long as a train arrives at the current station, passengers (the number of passengers should be no more than C) are allowed to board the train. For some oversaturated urban rail networks such as Beijing or Tokyo, however, some passengers may not be able to board the next arrival train, when the number of waiting passengers is more than the train's available capacity. Thus, these passengers have to wait for the following train. In such cases, train capacity constraints are very important to model more realistic passenger flows in an oversaturated urban rail network.

We first consider the explicit train capacity constraints at station i. Define $\mathcal{S}_i = \{1, 2, \dots, i-1\}$ as a set of stations before station i along the line. The above set indicates that there are passengers who start from station $i' \in \mathcal{S}_i$ and end their journey at station i. To calculate the total number of in-vehicle passengers when the next train arrives at station i, we generate a path set $\mathcal{PA}_{i',i}$ as follows:

$$\mathcal{PA}_{i',i} = \mathcal{PA}^o_{i'} \cap (\mathcal{PA}^d_{i+1} \cup \mathcal{PA}^d_{i+2} \cup \cdots \cup \mathcal{PA}^d_{|S|})$$

Each path in $\mathcal{PA}_{i',i}$ originates from the station i' and ends at stations that are located after station i. In other words, each path in $\mathcal{PA}_{i',i}$ crosses over station i.

According to the above definitions, we know the fact that the passengers with path $pa \in \mathcal{PA}_{i',i}$ are still on the corresponding train when this train departs from station i. We consider that the number of passengers with path $pa \in \mathcal{PA}_{i',i}$ is n_{pa}. Then, the available capacity is $C - \sum_{pa \in \mathcal{PA}_{i',i}} n_{pa}$. In other words, at most $C - \sum_{pa \in \mathcal{PA}_{i',i}} n_{pa}$ passengers can board the current train when this train departs from station i, and other passengers have to keep waiting for the next coming train. Therefore, we can formally derive the following train capacity constraint.

$$\sum_{pa \in \mathcal{PA}_i} PB^t_{pa,i} \le C - \sum_{i' \in \mathcal{S}_i} \sum_{pa \in \mathcal{PA}_{i',i}} PB^{t(i',i)}_{pa,i'}, \quad \forall i \in \mathcal{S},\ t \in \mathcal{T}_{pa,i} \tag{30}$$

where $t(i', i) = t - t_{i',i}$ indicates the passengers' boarding time at the former station $i' \in \mathcal{S}_i$, where $t_{i',i}$ is a parameter that represents the traveling time from station i' to i on the line. In constraint (31), the left term indicates the number of boarding passengers,while the right term indicates the available boarding capacity.

4.5. Extended formulation with dwelling time constraints

As urban rail lines operate in many cities, the accessibility and coverage of urban rail transit networks have increased significantly. This phenomenon indicates that the trend of network operation is possible. Currently, improving station services is a vital problem. The most intuitive reflection of station service level is passenger satisfaction. We notice that the generation of train timetables depends largely on the dwelling time, which is a particularly important factor in the quality of passenger service. Over the years, more research has been conducted on the train dwelling time, which is a key parameter for system performance and service reliability.

The train dwelling time is the time for a train to stand on the platform to complete its corresponding station work, frequently to allow passengers to board or alight. When alighting and boarding passengers are in low demand, the train dwelling time depends more on the train control strategy than on the number of boarding and alighting passengers (Seriani & Fernandez, 2015). However, as the passenger demand increases, the dwelling time becomes more dependent on the number of alighting and boarding passengers (Hunter-Zaworski, 2003). Thus, high alighting and boarding efficiencies are useful in reducing train dwelling time during peak hours. In a 2012 report, Douglas (2012) grouped the main factors influencing the dwelling time into six headings: passenger volume, passenger composition, train design, station design, timetable, and operational factors. San and Masirin (2016) indicated that passenger boarding and alighting behaviors are likely to be the most significant factors causing dwelling time variations. The dwelling time is determined by the number of passengers alighting and boarding and the speed at which they alight and board. The mixed flow of passengers tends to lengthen the dwelling time compared with a unidirectional flow. As early as 2006, Harris (2006) observed that the efficiency of passenger boarding and alighting on platforms is frequently closely related to passenger volume and demand as well as the on-vehicle crowding level (i.e., the ratio of the number of passengers in the train to the maximum train capacity).

Many studies related to the alighting and boarding processes have focused on predicting the dwelling time according to the estimated time of passengers boarding and alighting. In 2000, Puong (2000) proposed a train dwelling time model based on observations from the MBTA Red Line in Boston. They stated that the train dwelling time DT is a function of the number of passengers boarding and alighting (linear part) and the on-vehicle crowding level (nonlinear part), as follows:

$$DT = 12.22 + 2.27 \cdot B_d + 1.82 \cdot A_d + 6.2 \times 10^{-4} \cdot TS_d^3 B_d \tag{31}$$

where A_d and B_d refer to the number of alighting and boarding passengers, respectively, and TS_d is the number of standees per door on the train (i.e., the total number of standees divided by the number of doors). From (31), we observe that the alighting time is unaffected by the boarding passengers and the on-vehicle crowding level. This may be due to the fact that passengers obey the principle of "Alight first then get aboard" when boarding and alighting, without interfering with each other. Nevertheless, the dwelling time is sensitive to the flow of passengers onboard and on-vehicle congestion. As the number of passengers on the train gradually increases, the above two factors significantly influence the train dwelling time.

In the operation of urban rail transit, in addition to train punctuality, running speed, departure interval, and other indicators of high concern for passengers, train dwelling time practically affects passengers' travel experiences in an invisible manner. If the dwelling time is too long, the interval between train departures increases accordingly, and the traveling time of passengers is extended, resulting in a negative travel experience. In contrast, if the dwelling time is too short and the number of passengers boarding and alighting is too large, passengers cannot complete boarding and alighting the train within this time, which also causes dissatisfaction and complaints from passengers.

According to the dwelling time model developed by Puong, the train rescheduling model can be further optimized using passenger flow statistics and on-vehicle standees to achieve high resilience in urban rail transit systems.

The dwelling-time model suggests that the minimum dwelling time depends on the efficiency of passenger boarding and alighting and on-vehicle congestion. The three decision variables in the dwelling time model can be derived from our simulation of the passenger flow:

$$B_d = \Big(\sum_{\tau \in \mathcal{T}_t} \sum_{pa \in \mathcal{PA}^o_i} PB^\tau_{pa,i}\Big) / N_{door}, \quad \forall i \in \mathcal{S},\ t \in \mathcal{T} \tag{32}$$

$$A_d = (\sum_{i' \in S_i} \sum_{pa \in \mathcal{PA}_{i',i}} PB^{t(i',i)}_{pa,i'})/N_{door}, \quad \forall i \in S,\ t \in \mathcal{T}_{pa,i} \tag{33}$$

$$TS_d = (C - \sum_{i' \in S_i} \sum_{pa \in \mathcal{PA}_{i',i}} PB^{t(i',i)}_{pa,i'})/N_{door}, \quad \forall i \in S,\ t \in \mathcal{T}_{pa,i} \tag{34}$$

where N_{door} is the total number of doors; The specific meanings of the other polynomials are expressed in constraints (25)—(30). Subsequently, based on the train rescheduling model, a minimum dwelling time constraint (35) is added to limit constraint (7) considering passenger flows.

$$T^D_{e,\bar{e}} = 12.22 + 2.27 \cdot B_d + 1.82 \cdot A_d + 6.2 \times 10^{-4} \cdot TS_d^3 B_d \tag{35}$$

Finally, the disruption-management problem of train rescheduling in urban rail transit systems can be summarized in the following resilience-oriented model:

$$\begin{aligned}
\min \quad & (\max_i \frac{\sum_{\tau \in \mathcal{T}_{disrupt}} PW_i^\tau}{NT \cdot SC_i}), \\
& \forall i \in S,\ \mathcal{T}_{disrupt} = \{t \in \mathcal{T} \mid t_{start} \le t \le t_{end}\} \\
\text{s.t.} \quad & t^a_e - \bar{t}^a_e \ge 0, \quad \forall e \in \mathcal{E} \\
& t^a_e - \bar{t}^a_e \le (1 - c_e) * M, \quad \forall e \in \mathcal{E} \\
& constraints\ (3) - (20), (23), (24), \\
& (25) - (30), \\
& (32) - (35).
\end{aligned}$$

5. Solution methodology

In this section, a two-phase iterative optimization approach is developed to solve the MINLP problem, which is inspired by Chai, Yin, D'Ariano, Samà, and Tang (2023b). Phase 1 focuses on the adjustment of trains in case of disruptions, whereas phase 2 evaluates the impact of passengers on the adjusted train schedule. Based on these two phases, the proposed approach obtains a local optimum for the original MINLP problem. A flowchart of this approach is presented in Fig. 6. The first stage is to adjust the planned timetable according to the set parameters of disruptions (including the start time, duration and disrupted sections) with the principle of trade-off between train delays and service cancellations. In the second stage, the passenger simulation is developed to evaluate the rescheduled timetable. Results from the simulation provide the guidance for further rescheduling trains in the first stage.

Phase 1: Train Rescheduling

$$\begin{aligned}
\min_{e,e'} \quad & f = w_1 \sum_{e \in \mathcal{E}} (t^a_e - \bar{t}^a_e) + w_2 \sum_{e \in \mathcal{E}} c_e + w_3 \sum_{e' \in \mathcal{E}^T} (\alpha_{e'} + \beta_{e'}) \\
\text{s.t.} \quad & t^a_e - \bar{t}^a_e \ge 0, \quad \forall e \in \mathcal{E} \\
& t^a_e - \bar{t}^a_e \le (1 - c_e) * M, \quad \forall e \in \mathcal{E} \\
& (3) - (20), (23) - (24)
\end{aligned} \tag{36}$$

The first phase involves obtaining a trade-off between train delays and canceled services when rescheduling the rolling stock on the line. Proved by facts, the canceled service can have a direct impact on traveling passengers and may lead to the overcrowding of passengers at the platforms. The reserved trains can compensate for further delays caused by disruptions. With an increase in the number of reserved trains, operational costs also increase. Hence, a new objective function (36) is introduced in this phase, where w_1, w_2, and w_3 are weight coefficients representing the costs of train delays, canceled services, and rolling stock, respectively.

However, this MILP model contains four sets of binary variables and one set of continuous variables. Because of the significant number of dependent auxiliary variables, the model increases rapidly, making it difficult to generate an effective rescheduling timetable under a strict computational time limit under unexpected real-time disruptions. Furthermore, the introduction of big-M values can lead to a large duality gap between the root linear programming relaxation bound and the value of the feasible integer solutions. Therefore, a series of valid inequalities and the proper determination of big-M values are introduced into a branch-and-cut framework.

(1) Valid inequalities: To obtain tighter formulations for the model, we derive a series of valid inequalities before its resolution using a linear programming solver. Note that binary variables can be introduced to tighten constraints (9) and (10) as follows:

$$\sum_{e' \in D^+(e)} \lambda_{e,e'} \le 1 - c_e, \quad \forall e \in \mathcal{E}^{S^+} \cup \mathcal{E}^{T^+}, \tag{37}$$

$$\sum_{e' \in D^-(e)} \lambda_{e',e} \le 1 - c_e, \quad \forall e \in \mathcal{E}^{S^-} \cup \mathcal{E}^{T^-}. \tag{38}$$

It is clear that the above inequalities are valid for the model: If an event is canceled, i.e., $c_e = 1$, it cannot be connected to any other event, and $\sum_{e' \in D^+(e)} \lambda_{e,e'} = 0$.

To maintain operating safely, the following train is not permitted to pass through a station with a train dwelling. At the turnaround stations, the train tends to connect to the closest train that can be connected. In other words, any two events e and e' can be connected as short-turn operations if and only if the time between these two events is longer than the minimum turnaround time $T^T_{e,e'}$. Thus, we derive the following set of valid inequalities.

$$t^a_e + T^T_{e,e'} - t^d_{e'} \ge m * \Psi + \epsilon * (1 - \Psi(e)), \quad \forall e \in \mathcal{E}^{S^+} \tag{39}$$

where $\Psi(e) = \lambda_{e,e'} + c_{e'} + c_e$, m is the lower bound of $t^a_e + T^T_{e,e'} - t^d_{e'}$, e' represents the turnaround event of e given by the planned timetable, and ϵ is a small fractional value. Because $\Psi(e) = \lambda_{e,e'} + c_{e'} + c_e$, we notice that there are only three possible cases, i.e., $\Psi(e) = 0, 1, 2$ according to inequalities (37) and (38). First, if $\Psi(e) = 2$, both events e and e' are canceled in the rescheduled plan (i.e., $c_e = c_{e'} = 1$). Thus, we obtain $t^a_e = \bar{t}^a_e$, $t^d_{e'} = \bar{t}^d_{e'}$, and $m = t^a_e + T^T_{e,e'} - t^d_{e'} = 0$. The left-hand side of (39) equals 0, and the right hand of (39) is less than 0. Thus, inequality (39) always holds. Next, if $\Psi(e) = 1$, event e or event e' is canceled, and $\lambda_{e,e'} = 0$. In this case, the right-hand side of inequality (39) becomes m; thus, $t^a_e + T^T_{e,e'} - t^d_{e'} \ge m$ is always true according to the definition of m. Finally, if $\Psi(e) = 0$, both of events e and e' are not canceled but are not connected by turnaround operations (i.e., $c_e = c_{e'} = 0$ and $\lambda_{e,e'} = 0$). As $c_{e'} = 0$, we observe that there exists an event $\bar{e} \in \mathcal{E}^{S^+}$ connects to event e'. According to constraint (14), we obtain the following inequality:

$$t^a_e \ge t^d_{e'} + T^P_{e',e} \tag{40}$$

As $T^P_{e',e} \ge 0$, we obtain $t^a_e > t^d_{e'}$. Subsequently, the left-hand side of (39) can be reformulated as

$$t^a_e + T^T_{e,e'} - t^d_{e'} > T_{depot} > 0, \tag{41}$$

which indicates that the inequality in (39) is always true if $\Psi(e) = 0$.

(2) Determination of big-M values: In the following, we select proper values of M correctly to avoid the integrality tolerance problem and producing sufficiently tight bounds. (i) Regarding M in constraint (7), we can set the maximum dwelling time of each event e as the disruption duration time $t_{end} - t_{start}$ to appropriately tighten the constraint. (ii) Note that $t^d_{e'} \ge t^d_e$ if $e' \in H^+(e)$ (i.e., event e' follows event e). Hence, the value of M in constraint (8) can be set to $T^H_{e,e'}$. (iii) For the big-M values in (11), (14), (15), (22), and (23), their values should be at least equal to the latest end of the last event's arrival time t^{latest}_e, which is given by

$$t^{latest}_e \ge t_{end} - t_{start} + \max_{e,e' \in \mathcal{E}} \{|t^a_e - t^d_{e'}|\} \tag{42}$$

Phase 2: Evaluation of the Adjusted Train Schedule Although the model above accomplishes train rescheduling after disruptions, the adjusted train schedule obtained from the model does not necessarily contribute to an improvement in system resilience. Thus, the evaluation of adjusted schedules is indispensable for achieving the objective

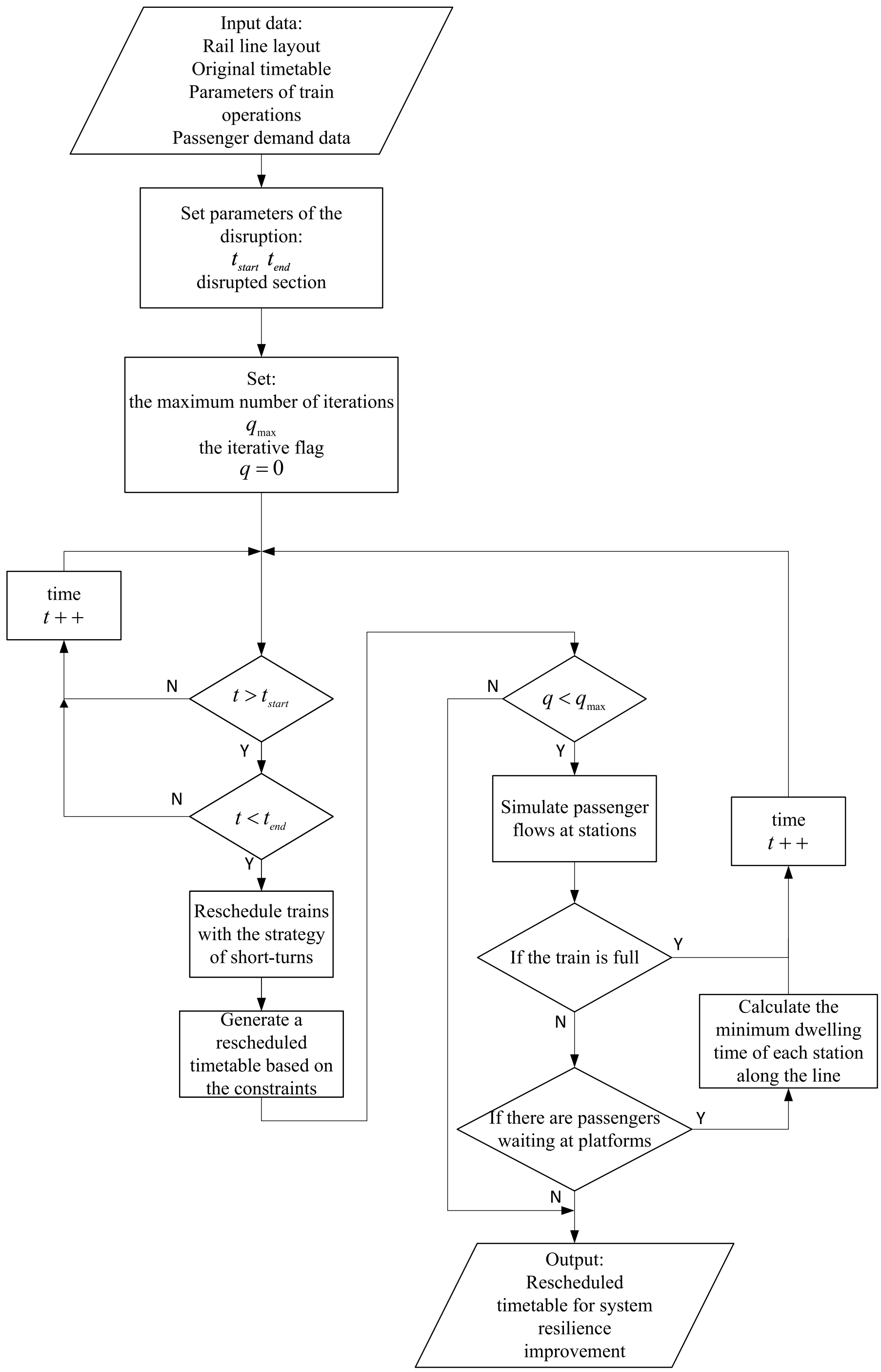


Fig. 6. Two-phase iterative optimization approach.

of enhancing system resilience. The detailed evaluation procedure is presented in Algorithm 1.

Taking Beijing Metro Line 1 as an example, in this algorithm, we first use the existing timetable of Line 1 as the initial input and set some parameters for the disruption, such as the time at which the disruption occurs, the affected interval, and the duration. Subsequently, the train rescheduling model can generate an initial solution, which is used as the input to the simulation model of coupled passenger-train flows, combined with historical passenger flow data, to determine the number of stranded passengers at each time point and station along the line during the operation. As the termination decision of this algorithm, Step 3 determines whether a scenario exists in which the dwelling time is too short to board on the train for some passengers. If no such scenario exists, the program is expected to be terminated because the dwelling time is sufficient to satisfy the passenger boarding and alighting demands. In other words, the current train schedule is optimal. Otherwise, step 3.2 proceeds to discuss whether the reason for passengers being stranded on platforms is inadequate train capacity or a short dwelling time. The former is determined by the properties of the rolling stock, and we cannot solve the problem by adjusting the current timetable. For the latter, step 3.2.2.1 calculates the minimum dwelling time according to constraint (35). With these new parameters, the train rescheduling model regenerates the adjusted schedule and repeats the steps above until the end of the iterative loop to obtain an optimal timetable.

Algorithm 1: The evaluating procedure for resilience improvement

1: **Input**: corresponding parameters of the adjusted train schedule, i.e., the train arrival time t_e^a and departure time t_e^d of each station on upstream and downstream and dwelling time at each station, time-dependent passenger dynamic OD matrix based on historical data, passenger boarding rate, maximum train capacity C, maximum number of iterations q_{max};
2: iteration index $q \leftarrow 0$;
3: **Repeat**
4: Begin the simulation, $q = q + 1$;
5: Whether there are passengers stranded at the station after the current train has left the station:
5.1: If yes, go to step 6;
5.2: Otherwise, go to step 10;
6: Whether the current train has spare capacity for boarding passengers:
6.1: If yes, calculate the minimum dwelling time $\hat{T}^D_{e,\bar{e}}$ at the current station by Eqs. (35);
6.2: Otherwise, go to step 10;
7: Compare the values of output dwelling time $T^D_{e,\bar{e}}$ and calculating dwelling time $\hat{T}^D_{e,\bar{e}}$:
7.1: If $\hat{T}^D_{e,\bar{e}} > T^D_{e,\bar{e}}$, update the output dwelling time by the calculating values;
7.2: Otherwise, keep the output values;
8: Input the calculated new timetable into the train rescheduling model;
9: **Until** there are no more updates on train dwelling time or $q = q_{max}$
10: **Return** a resilience-oriented rescheduling timetable

Table 4
Parameter settings of the illustrative example.

Description	Value
Number of stations	8
Running times in the up direction	120, 143, 120, 85, 140, 120, 200
Running times in the down direction	200, 120, 140, 85, 120, 143, 140
Dwelling times in the up direction	0, 30, 45, 45, 30, 35, 25, 0
Dwelling times in the down direction	0, 25, 30, 30, 45, 45, 30, 0
Station capacity	2000

6. Numerical experiments

In this section, we present the computational results on a small realistic case study and a real-world case study to illustrate the potential application and advantages of the proposed models and solution approaches. In the first set of experiments, we respectively use heuristic method, benchmark CPLEX solver, and two-phase iterative optimization approach —hereafter termed as HEM, CPLEX, and TPIO respectively. We compare the performances of these three approaches in terms of solving quality, resilience indicators and computational time. In the second set of experiments, we conduct sensitivity analysis to demonstrate impacts of the weight factor w_2 (i.e., adjustment strategies) and operation periods in system resilience. We evaluate the improvement of system resilience during both of peak and off-peak periods based on the historical passenger flow data of Beijing Metro Line 1. All the experiments are coded in Visual C++ 2017 on a 64-bit computer with Intel(R) Core(TM) i7-8550U CPU @ 1.80 GHz 2.00 GHz and 8.00 GB RAM. And we apply the state-of-the-art IBM ILOG CPLEX Optimization Studio 12.8 solver to solve the proposed model.

6.1. Small case study

6.1.1. Experiment description and parameter settings

The illustrative example selects a typical bidirectional metro line with eight stations and one depot connected to station 1 for simplification. For simplicity, in this example, we assume that the capacity of each station is the same, and we can thus set the station capacity to 1 for convenience. Disruption is considered to occur between stations 4 and 5 from time $t_{start} = 2500$ s to $t_{end} = 5500$ s. The basic parameters are presented in Table 4.

In this illustrative example, we generate a total of ten instances with different values of parameter w_2 during peak and off-peak hours. The weight coefficient w_2 represents service cancellations, which can affect the generation of the adjustment strategies. The research objective of this paper is to enhance the resilience of urban rail systems, and the resilience metric, i.e., the average congestion of the busiest station during operational time, has been introduced in Section 3.1, depending on the adjustment strategies adopted. To simulate peak and off-peak hours, we randomly generate the time-dependent passenger demand as follows: For the peak hours, a monotonically increasing parameter is used to indicate the passenger arrival rate, whereas for the off-peak hours, a monotonically decreasing passenger arrival rate is generated. We then randomly generate OD pairs according to the passenger arrival rate. For comparison, we adopt three methods to generate a train timetable in the case of disruption. The first is a heuristic method (termed as HEM) that holds all trains on the line until the disruption is resolved. Once the disruption occurs, trains, which subsequently serving the disrupted section, stop at the platform closest to it, and the affected train service will not continue until the disruption is fully restored. The second benchmark employs a standard CPLEX solver with the default settings to solve our rescheduling model. The third is our two-phase iterative optimization approach with valid inequalities (termed as TPIO).

6.1.2. Comparison among various solution methods

Table 5 presents the results of the HEM, CPLEX and TPIO methods for solving the instances. Columns 4—7 show the resilience metric value, total train time delay, number of canceled events, and computational time, respectively. We can observe from the results that the heuristic HEM achieves the worst results compared with both CPLEX and TPIO for all instances. HEM simply “freezes” trains such that they cannot continue transporting passengers during disruptions, leading to a gradual accumulation of passengers on the platforms. This phenomenon causes severe congestion at the stations and reflects a very low level of system resilience. Although some events are canceled, CPLEX and TPIO clearly outperform HEM, indicating that our MINLP model can enhance system resilience to disruptions. The short-turning strategy in non-disruptive sections satisfies the travel demands of some passengers to relieve the load pressure on the stations. As Table 5 shows, the most significant difference between CPLEX and TPIO is the computational time. The computational time is clearly reduced by TPIO compared with that of CPLEX, which supports that the application of our approach is realistic.

The experimental results of this small case study demonstrate that our TPIO approach significantly reduce train delays and provide more travel services for passengers compared to the traditional heuristic method in case of disruptions. Although some affected train services are canceled, TPIO approach continues providing services to passengers by short-turning strategies during the disruption period. Moreover, the acceptable calculation time shows the possibility for the application of TPIO approach in practical scenarios.

Table 5
Computational results by HEM, CPLEX and TPIO for different instances.

Time period	Value of w_2	Method	Resilience $\mathcal{R}$	Total delay time	Total canceled events	Computational time (s)
Peak hours	10	HEM	0.9585	523 238	0	0.074
		CPLEX	0.8109	780	32	140.879
		TPIO	0.8109	780	32	43.127
	100	HEM	0.9585	523 238	0	0.071
		CPLEX	0.7899	949	20	53.396
		TPIO	0.7899	949	20	15.543
	200	HEM	0.9585	523 238	0	0.07
		CPLEX	0.7899	949	20	50.335
		TPIO	0.7899	949	20	11.196
	300	HEM	0.9585	523 238	0	0.078
		CPLEX	0.7850	2686	14	45.392
		TPIO	0.7850	2686	14	11.901
	500	HEM	0.9585	523 238	0	0.072
		CPLEX	0.7850	2686	14	48.113
		TPIO	0.7850	2686	14	14.758
Off-peak hours	10	HEM	0.062	523 238	0	0.074
		CPLEX	0.0572	793	32	58.88
		TPIO	0.0572	793	32	21.562
	100	HEM	0.062	523 238	0	0.071
		CPLEX	0.0572	968	20	29.781
		TPIO	0.0572	968	20	7.715
	200	HEM	0.062	523 238	0	0.07
		CPLEX	0.0572	1266	16	25.941
		TPIO	0.0572	1266	16	5.597
	300	HEM	0.062	523 238	0	0.078
		CPLEX	0.058	2195	14	25.415
		TPIO	0.058	2195	14	5.854
	500	HEM	0.062	523 238	0	0.072
		CPLEX	0.058	2604	10	28.136
		TPIO	0.058	2604	10	7.431

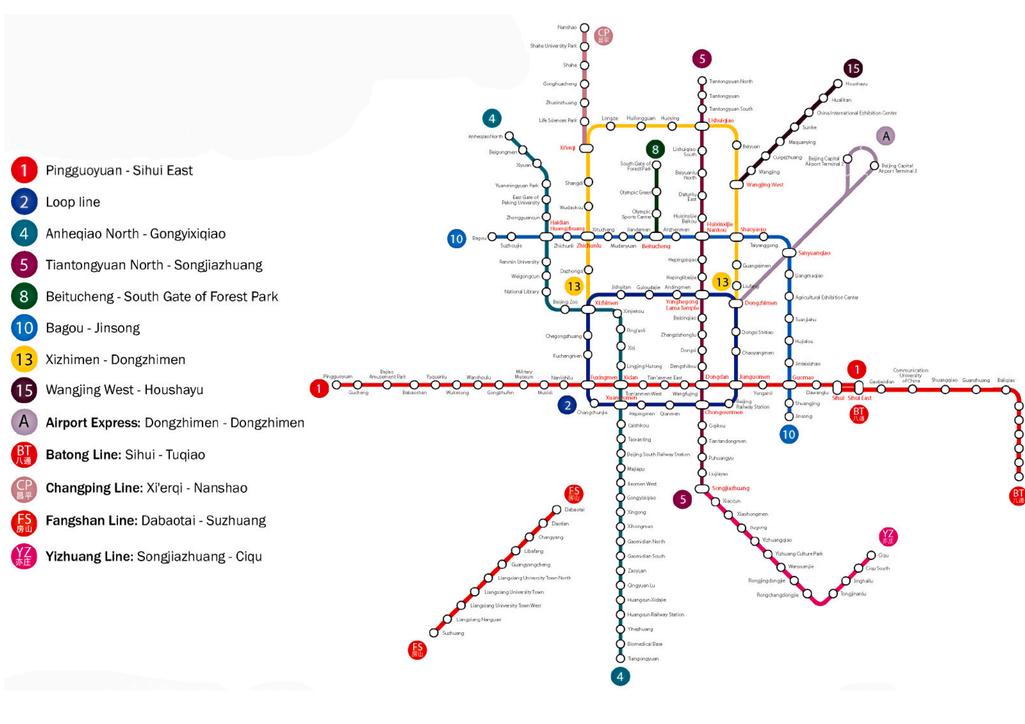
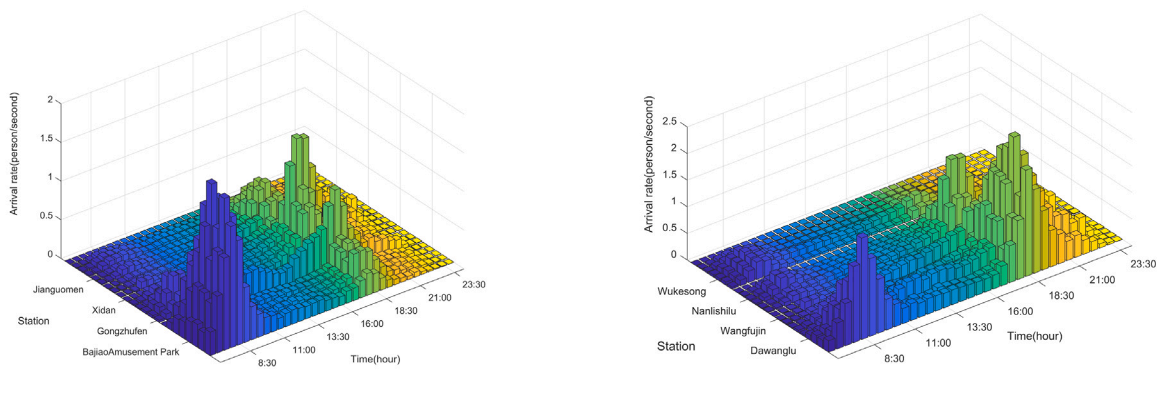


Fig. 7. The layout of Beijing Metro Line 1.

Table 6
Basic parameters of Beijing Metro Line One.

Description	Value
Number of stations	23
Number of depots	2
Minimum headway time	427 s
Minimum turnaround time	145 s
Planned running time of each upstream section	200,150,150,120,140,135,110,100,100,120,60,120, 105,90,90,85,110,120,85,120,143,120,0 s
Planned running time of each downstream section	0,140,143,120,85,120,110,85,90,90,105,120,80,120, 100,100,110,135,140,120,150,150,200 s
Planned dwelling time of each upstream station	30,30,30,25,35,40,35,45,50,30,27,50,44,26,28,30,45, 45,30,45,45,30,30 s
Planned dwelling time of each downstream station	30,30,30,25,30,35,35,41,50,30,26,50,45,26,26,30,45, 45,30,45,45,30,30 s



(a) Line 1 up direction (b) Line 1 down direction

Fig. 8. Passenger arrival rate of Beijing Metro Line 1.

6.2. Real-world case study on Beijing Metro Line One

6.2.1. Experiment description and parameter settings

We use the real-world data from Beijing Metro Line 1 to investigate the impact of train rescheduling decisions on the system resilience. As the first underground rail line in China, Beijing Metro Line 1 covers a total of 23 stations and 22 sections that crosses the downtown of Beijing, as depicted in Fig. 7. These 23 stations are numbered in the up-direction and down-direction respectively. We group the stations in the up-direction as a set $\overline{S} = \{1, 2, 3, \ldots, 23\}$ and the stations in the down-direction are denoted by a set $\underline{S} = \{24, 25, 26, \ldots, 46\}$. The stations can accommodate no more than 2500 passengers. In the daily operation, the planned train number is 59 and the total operational service number is 798. In this operational, the headway is set $T^H_{e,e'} = 427$ s and the turnaround time is $T^T_{e,e'} = 145$ s. The other operational data, such as dwelling time at each station and running time on each section, are appended in Table 6.

In the case study, it is assumed that the disruption occurs between station 14 and station 15 (i.e., Muxidi-Military Museum) within the time $[t_{start}, t_{end}] = [7000\text{ s}, 9000\text{ s}]$. When the train runs to the disrupted section, the only measure to avoid the disrupted section is the short-turning strategy; otherwise, it has to dwell at the station until normal operation is resumed. Beijing Metro Line 1 adopts rolling stock consisting of six B Type Metro trains that can accommodate up to 1440 passengers at the same time. Dynamic passenger flow data comes from history Automatic Fare Collection (AFC) data collected on January 6, 2020, a graph of the change in passenger arrival rates in the upward and downward directions is shown in Fig. 8.

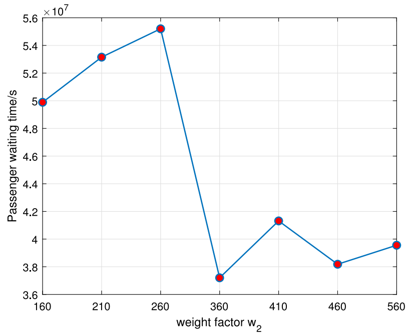
6.2.2. Comparison of different adjustment strategies

In the objective function of the train rescheduling model (Eq. (2)), the weight factors w_1, w_2, and w_3 represent the contributions of the total delay, canceled services, and reserved trains, respectively. Specifically, the smaller the weight factor of canceled services, the more passenger services are canceled. In other words, the train tends to dwell at the station until normal operation is restored. This instance does not consider the scenario of reserved trains; therefore, the weight factor w_3 is set to 0. Moreover, we set $w_1 = 1$ to impose a penalty of 1 on the objective function when the train is delayed for 1 s. To investigate the impact of various strategies on system resilience, we set the value of the weight factor w_2 to 160, 210, 260, 360, 410, 460, and 560, respectively. The rescheduling model generates different timetables using various strategies, and the results obtained from the passenger flow simulation are shown in Fig. 9.

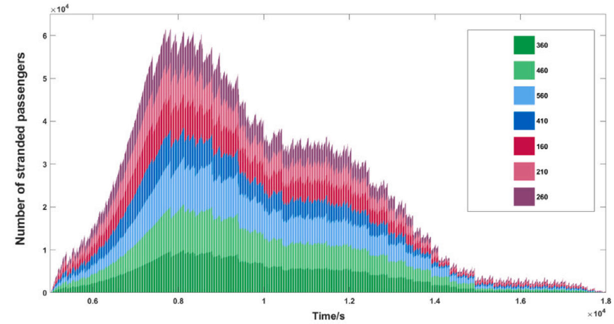
Figs. 9(a) and 9(b) show the variation in the two indicators of total waiting time and total number of stranded passengers at the stations when we increase w_2. The results indicate that both indicators perform the best (i.e., the shortest total waiting time and lowest total number of stranded passengers) when w_2 is set to 360. More significantly, both indicators exhibit a sharp decrease when w_2 increases from 260 to 360. This phenomenon indicates that the rational use of track resources for short-turning operations can not only better serve passengers whose travel is affected by the disruption but also satisfy the travel demands of passengers at non-disrupted stations. Nevertheless, an appropriate adjustment strategy for trains must be as balanced as possible, considering the limited track resources and the impact of associated train delays owing to increased passenger services.

6.2.3. Impact of congestion level on train delays

In this experiment, we also consider the scenario in which overcrowded stations in actual operation may exacerbate train delays. The initial timetable generated by the existing train rescheduling model maintains the dwelling times of each station in the planned timetable, whereas the adjusted timetable after iterative optimization adjusts

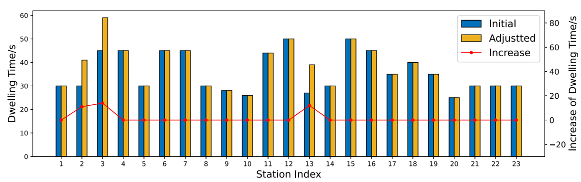


(a) Passenger waiting time with weight factor w_2

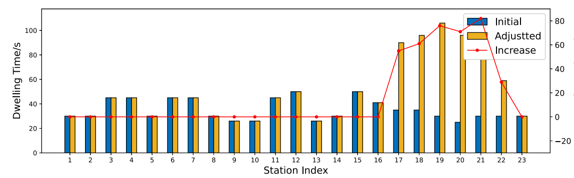


(b) Number of stranded passengers with weight factor w_2

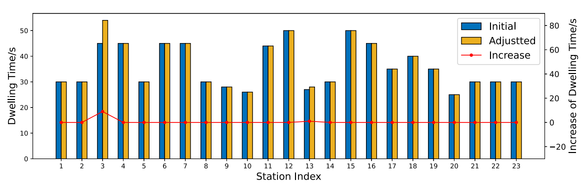
Fig. 9. Impact of different adjustment strategies on passengers.



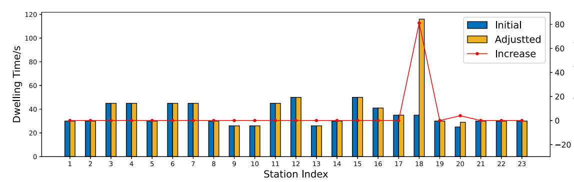
(a) Up-direction(8:00-9:30)



(b) Down-direction(8:00-9:30)



(c) Up-direction(10:30-12:00)



(d) Down-direction(10:30-12:00)

Fig. 10. Dwelling time with adjustment.

the dwelling times according to time-dependent passenger demands. Based on the historical data of passenger travel, we conduct disruption experiments during peak and off-peak hours, and a comparison of train dwelling time at each station before and after adjustment is shown in Fig. 10. Figs. 10(a) and 10(b) show the statistical results for the peak hours. We observe a total of 10 stations with extended dwelling times. The dwelling times at stations 17, 18, 19, 20, 21, and 22 have been extended to more than 20 s, owing to the large number of boarding passengers. Additionally, Figs. 10(c) and 10(d) show the results for off-peak hours. Compared with the variations in peak hours, only three stations increase the dwelling time during off-peak hours because of the overall lower intensity of passenger flows. The main task of trains at a station is passenger boarding and alighting operations. In overcrowded stations, trains must spend more time at stations to accomplish these tasks, which further aggravates train delays. Thus, the impact of passenger overcrowding on train operations on platforms cannot be ignored.

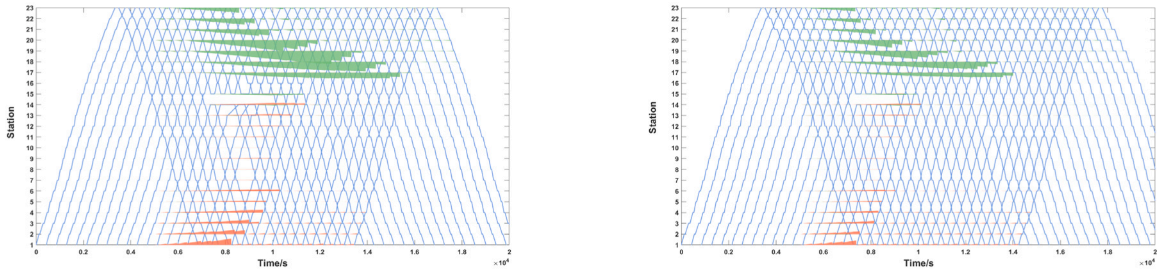
6.2.4. Evaluation of resilience improvement

In the previous section, we determined an appropriate adjustment strategy (i.e., $w_2 = 360$) and then separately evaluated our proposed model for resilience improvement during peak and off-peak hours.

Evaluation in peak hours. Figs. 11(a) and 11(b) show the train schedules before and after the adjustment during the peak hours, respectively. The number of accumulated passengers at all involved stations is also shown in Fig. 11, where the passenger accumulation curve in the downward direction is shown by the bars under the horizontal line of each station, and the passenger accumulation curve in the upward direction is indicated by bars above the horizontal line. Fig. 11 shows clearly that the number of passengers accumulated at each platform increases very rapidly when the disruption occurs. In contrast, the optimal schedule increases the number of train services to satisfy more passenger travel demands, and the number of stranded passengers at each station decreases significantly.

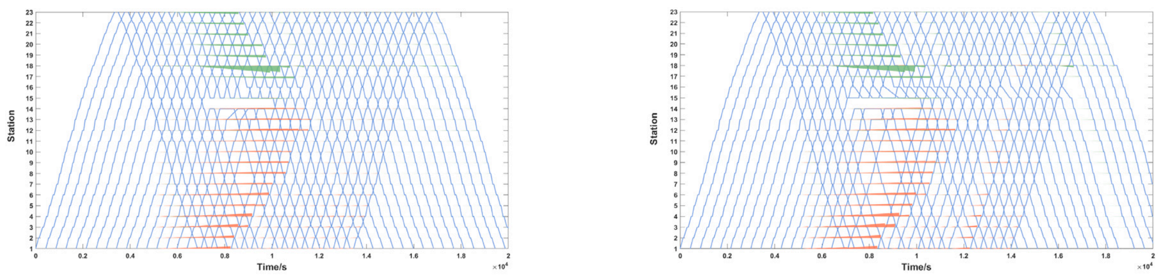
Table 7 shows the specific computation results for the peak hours. The results indicate that our train rescheduling model improves the system resilience by 39.7% and decreases the number of passengers accumulated at all stations by 21,436,037. The decrease in passenger flow density at the stations can significantly contribute to reducing the probability of virus transmission. Owing to the iterative process, the computation time of our approach is considerably longer than that of common rescheduling models. The proposed model can enhance service performance during disruptions to a certain extent.

Evaluation in off-peak hours. Similar to Fig. 11, Figs. 12(a) and 12(b) show the train schedules before and after the adjustment in off-peak hours, respectively. Comparing Figs. 11 and 12, we observe that the effect of optimization on reducing stranded passengers is not apparent because the intensity of passenger flow is low during off-peak hours, and the initial dwelling time of trains can basically satisfy passenger boarding and alighting demands. From the results in Table 8, during off-peak hours, the system resilience is improved by only 7.4%, and the number of passengers accumulated at all involved stations is reduced by 68,258 after optimization. In this adjustment, only three stations had extended dwelling times, and the adjusted timetable is also computed faster than during the peak hours. We also observed that the number of stations with congestion is not high because of the lower passenger demand during off-peak hours. With minimal passenger impact, train delays can be eliminated in a relatively short period and can return to running as planned. This phenomenon indicates that our proposed



(a) Initial timetable   (b) Adjusted timetable

Fig. 11. Simulation of Beijing Metro Line 1(8:00–9:30).



(a) Initial timetable   (b) Adjusted timetable

Fig. 12. Simulation of Beijing Metro Line 1(10:30–12:00).

Table 7
Computation results in peak hours.

	R	Number of waiting passengers	Computational time	Number of stations with adjusted dwelling time	Improvement of resilience level
Initial timetable	0.1556	51 687 822	2311.78 s	10	39.7%
Adjusted timetable	0.0938	30 251 785	8102.27 s		

Table 8
Computation results in off-peak hours.

	R	Number of waiting passengers	Computational time	Number of stations with adjusted dwelling time	Improvement of resilience level
Initial timetable	0.0592	24 754 682	2311.78 s	3	7.4%
Adjusted timetable	0.0548	24 686 424	4635.27 s		

approach has a significant effect on resilience improvement in overcrowded conditions, whereas the method of resilience improvement in normal passenger flows must be explored.

7. Conclusions and future works

This paper addresses the problem of resilience enhancement of urban rail transit systems considering overcrowded platforms in daily operation scenarios, focusing on the impact of disruptions on passenger travel. Although this paper studies only the Beijing Metro System as an example, the proposed models can be applied to other urban rail systems worldwide. The major conclusions are summarized as follows:

1. A train rescheduling optimization is formulated for the resilience enhancement of urban rail systems by considering the congestion of stations. To solve this MINLP model, we present a two-phase iterative optimization approach. Phase 1 focuses on the adjustment of trains to enhance the system resilience, and phase 2 evaluates the impact of passengers on the adjusted train schedule. The result indicates that the train rescheduling model can effectively optimize the system's resilience. The proposed train rescheduling model can aid dispatchers in selecting the optimal timetable in real-time rail traffic managements.
2. The resilience indicator, which is the congestion level of the busiest station, provides a more intuitive measurement of station service level than topology and operational performance. Whether considering the passengers' safety or the epidemic prevention requirements in public places, dispatchers should promptly pay attention to the congestion scenario of stations and coordinate the requirements of operation and passenger service when rescheduling trains. Relatively extending the dwelling time at congested stations can not only significantly reduce the capacity pressure of stations but also increase the transportation capacity of systems.
3. The resilience of urban rail systems varies significantly under the influence of time-varying passenger flows. In particular, disturbances are more harmful to the system in peak hours than off-peak hours. Ensuring the normal operations of trains at busy stations with high passenger density has practical meaning for avoiding highly negative impacts on urban rail systems during peak hours.

This study has four limitations. (1) The passenger travel paths are assumed to be predetermined using historical data from operating metro companies. Nevertheless, passenger behavior, in practice, is complex and random. Passengers may also select different traveling paths based on the actual train timetables. In addition, it is common for some passengers to transfer to other lines or change their traveling patterns because of station congestion and other reasons. Therefore, the traveling behavior of passengers should be analyzed to consider these uncertainties in the future. (2) The study object we selected is a single line of the urban rail transit system; thus, the transfer passenger flow is not considered in the passenger flow simulation or in the calculation of the minimum dwelling time of trains. A critical topic for extension of the present study is to consider the operational planning for a more complex rail network with multiple routes and with different origins and destinations. (3) As a large and complex system, the resilience of urban rail transit is influenced by various factors. In addition, the definition of resilience for different scenarios and lines will assume different forms depending on the demand. Overall, the quantitative evaluation of resilience was promising. (4) Another interesting direction for future research is considering integrated and connected multimodal transport systems included buses, taxis and shared-bikes. Multimodal transport systems provide passengers with additional travel options. When one traveling mode in the system is disrupted, other modes can be used as alternatives to reach the destination, which undoubtedly improves the overall resilience of the system.

CRediT authorship contribution statement

Zhixian Li: Study conception and design, Data collection, Analysis and interpretation of results, Draft manuscript preparation. **Jiateng Yin:** Study conception and design, Data collection, Analysis and interpretation of results, Draft manuscript preparation. **Simin Chai:** Study methodology and design, Data collection, Draft manuscript preparation. **Tao Tang:** Study conception and design, Data collection, Draft manuscript preparation. **Lixing Yang:** Draft manuscript preparation.

Data availability

Data will be made available on request.

Acknowledgments

This research was supported by the Fundamental Research Funds for the Central Universities (No. 2022YJS115), National Natural Science Foundation of China (Nos. 72322022, 72288101) and Beijing Natural Science Foundation (No. 4222051).

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